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Multi-view structural twin support vector machine with the consensus and complementarity principles and its safe screening rules

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ABSTRACT

Nowadays, numerous multi-view algorithms are proposed to achieve better performance in classification tasks. In this article, we propose a multi-view structural twin support vector machine (MvSTSVM), which is guided by both complementarity and consensus principles. In order to keep the consensus information, structural information of inter-class and intra-class is explored in two hyperplanes by clustering methods, which significantly enhances the classification performance. Besides, MvSTSVM introduces the weights to multi-view datasets to dig the complementarity information among different views. In addition, our model achieves both intra-class compactness and inter-class separability. To deal with the optimization problems, an iterative strategy is adopted for the solution of MvSTSVM. Moreover, we derive the safe screening rules for MvSTSVM, which speed up the tuning process obviously without sacrificing accuracy. A series of numerical experiments confirm the effectiveness of MvSTSVM and the safety of proposed screening rules.

1. Introduction

In machine learning, for the purpose of classifying or identifying the samples efficiently, pre-feature extraction is usually performed on the datasets. With the diverse development of information acquisition technology, the same sample can be represented as feature data across distinct domains or distinct levels, which is known as multi-view data (Wang, Zhu, & Zhang, 2023). It possesses some characteristics such as polymorphism, high-dimensional, heterogeneity, multi-source and multi-description (Zong, Zhang, Zhao, Yu, & Zhao, 2017). For example, in the field of image processing, features like Gabor features, color histograms (CH) features and histogram of oriented gradients (HOG) features make up three views of an image, with each view containing unique information. Due to the distinct properties of different features, combining data from multiple views into single-view data fuzzy the characteristic information of the original perspective. What is worse, the increase in feature vector dimensions also raises the computational complexity of the model. Therefore, to improve the learning efficiency, it is more appropriate to build classifiers separately on each perspective, which is called multi-view learning (MVL). Compared with single-view learning, multi-view learning tends to achieve higher accuracy.

Existing MVL algorithms are mainly categorized into co-regularization algorithm, co-training algorithm and margin-consistency algorithm (Zhao, Xie, Xu, & Sun, 2017). The co-regularization algorithm integrates the regularization terms into the objective function to ensure consistency between different views. Typical models include SVM-2K (Farquhar, Hardoon, Meng, Shawe-Taylor, & Szedmak, 2005), multi-view twin SVM (MvTSVM) (Xie & Sun, 2015), multi-view SVM with the consensus and complementarity information (MvSVM2C) (Xie & Sun, 2019), multi-view learning with privileged weighted twin SVM (MPWTSVM) (Xu & Wang, 2022) and sub-views learning for PSVM-2V (SL-PSVM-2V) (Hao, Zheng, Xiao, & Zhu, 2023). Co-training algorithm is an important method in semi-supervised learning, which utilizes complementarity information between multiple perspectives to alternately learn and explore the attributes of unlabeled data. The algorithm includes three main steps: view acquisition, learner differentiation and label confidence estimation. It is effectively integrated into most existing machine learning algorithms (Ning et al., 2023). In addition, the margin-consistency algorithm based on maximum entropy discrimination (MED) (Sun & Chao, 2013) is employed to classify multi-view data, such as alternative multi-view MED (AMvMED) (Chao & Sun, 2015), consensus and complementarity-based MED (MED-2C) (Chao & Sun, 2016).

Despite differences among the current multi-view algorithms, they all follow the principle of complementarity or consensus. The consensus principle suggests that there are inner connections between features from different views of the same object, aiming to maximize

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the consistency among multiple perspectives. The complementarity principle means that information from distinct views can supplement each other. In this case, data can be described more comprehensively and accurately (Zhao & Xu, 2023).

Based on these two principles, many excellent multi-view algorithms have been proposed. For example, SVM-2K inherits the maximum margin of SVM and introduces the second view into the model. Its consistency is realized by the similarity constraint between two views. Farquhar et al. (2005) demonstrate that the Rademacher complexity of the corresponding class achieves a significant reduction. To improve the performance of SVM-2K, numerous multi-view classifiers with nonparallel hyperplanes are proposed, which obtain two nonparallel hyperplanes for each view. Multi-view twin SVM (MvTSVM) retains the similarity constraints in SVM-2K and derives two hyperplanes by solving two small-scale QPPs. Multi-view generalized eigenvalue proximal SVM (MvGSVM) and multi-view improved GEPSVM (MvIGSVM) (Sun, Xie, & Dong, 2018) obtain two hyperplanes in the manner of solving two generalized eigenvalue problems and effectively combine the two perspectives by maximizing the co-regularization term. These two models only satisfy the principle of consensus. Fortunately, there are many models simultaneously satisfy consensus and complementarity, such as multi-view TSVM with the consensus and complementarity information (MvTSVM2C) (Xie & Sun, 2019), multi-view structural large margin classifier (MvSLMC) (Zhao & Xu, 2023), multi-view large margin distribution machine (MvLDM) (Hu, Xiao, Zheng, Zhu, & Hsu, 2024) as well as multi-view universum SVM with insensitive pinball loss (MvUTSVM) (Lou & Xie, 2024), which greatly improves the classification performance of the model.

Although multi-view support vector machines can effectively mine the potential information between different perspectives, their complexity significantly surpasses single-view support vector machines. For example, suppose n_1, n_2 stand for the sample numbers of positive and negative classes respectively. Now we let $l = n_1 = n_2$, the complexity of nonparallel SVM (NPSVM) (Tian, Qi, Ju, Shi, & Liu, 2014) is $2\times\mathcal{O}((\frac{3}{2}l)^3)$, but the complexity of multi-view nonparallel SVM (MvNPSVM) (Tang, Li, Tian, & Liu, 2018) is inflated to $2\times\mathcal{O}((4l)^3)$, greatly affecting the processing speed of large-scale multi-view data. To accelerate SVM effectively, Ogawa, Suzuki, and Takeuchi (2013) propose that non-support vectors in SVM can be recognized before training and eliminated from the training samples in advance, which significantly reduces the scale of QPPs thereby speeding up the parameter optimization process. Since screening non-support vectors does not affect the solution of the classification hyperplane, this method is considered "safe" and is known as "safe screening rules" (SSR). Subsequently, Wang, Wonka, and Ye (2014) improve SSR by the approach of variational inequality (DVI). Recently, SSR has been used to speed up the optimization of multi-view support vector machine. For instance, based on the optimal conditions of SVM-2K and MvTSVM, Wang et al. (2023) firstly derive SSR-SVM-2K and SSR-MvTwSVM. Zhao and Xu (2023) introduce a MvSLMC and show SSR-MvSLMC at the same time. Additionally, SSR has been extended to other areas, including logistic regression (Yuan & Xu, 2023), zeros recognition (Elvira & Herzet, 2023), multi-label learning (Zhong & Xu, 2023), multitask learning (An, Xu, & Liu, 2023) and deep learning (Zeqiu Chen and Kaiyi Zhao and Shulin Sun and Jiayao Li and Shufan Wang and Ruizhi Sun, 2024).

In this article, motivated by the above research, we propose a novel multi-view structural twin support vector machine (MvSTSVM). It extends the structural twin support vector machine (S-TSVM) and inherits a series of advantages of MvTSVM, which brings new prospects for MVL. For computational efficiency, MvSTSVM acquires two hyperplanes by solving two QPPs. The first QPP aims to make the positive hyperplane from different views closest to the corresponding positive class and as far away from the negative one, while the second is the opposite.

The novelty of our model can be highlighted as the following remarkable characteristics:

(1) MvSTSVM can solve general multi-view classification problems, which is beneficial to dig the information of multi-view datasets.

(2) Our proposed method incorporates the idea of view weights into the object function, thus mining the potential similarity information among different views and realizing complementarity.

(3) MvSTSVM adopts the clustering technique to obtain structural regularization terms and collect the prior information within classes, which satisfies consensus in a non-pairwise way, thereby enhancing the model's classification efficiency.

(4) We introduce an iterative strategy into MvSTSVM to solve the optimization problems.

(5) We derive SSR-MvSTSVM based on KKT conditions and the variational inequality to reduce the scale of QPPs and speed up the tuning process.

The rest of this article is arranged as follows. In Section 2, we briefly review S-TSVM, TSVM and MvTSVM. Section 3 discusses the details of MvSTSVM. Section 4 presents the safe screening rules for MvSTSVM. In Section 5, we compare our model with several benchmark algorithms on two-view and multi-view datasets. In Section 6, we conclude our article and outline the future work.

2. Related works

In this part, we briefly retrospect TSVM (Khemchandani, Chandra, et al., 2007), S-TSVM (Qi, Tian, & Shi, 2013) and MvTSVM (Xie & Sun, 2015).

2.1. TSVM

For a binary classification problem, two matrices $\tilde{\boldsymbol{X}}_+$ and $\tilde{\boldsymbol{X}}_-$ are given, representing the positive dataset and the negative dataset respectively. $\tilde{\boldsymbol{X}}_+$ is a matrix of size $n_1\times p$ and matrix $\tilde{\boldsymbol{X}}_-$ is $n_2\times p$. TSVM aims to find two nonparallel hyperplanes:

$$f_+ : \langle \boldsymbol{u}_+, \boldsymbol{x}\rangle + b_+ = 0 \quad \text{and} \quad f_- : \langle \boldsymbol{u}_-, \boldsymbol{x}\rangle + b_- = 0, \tag{1}$$

where $\boldsymbol{u}_+, \boldsymbol{u}_- \in \mathbb{R}^p$, $\boldsymbol{x} \in \mathbb{R}^p$, $b_+, b_- \in \mathbb{R}$. Each hyperplane is close to the class it belongs to and far away from another. Now, we define $\boldsymbol{X}_+ = [\tilde{\boldsymbol{X}}_+ \ \boldsymbol{e}_+]$, $\boldsymbol{X}_- = [\tilde{\boldsymbol{X}}_- \ \boldsymbol{e}_-]$, $\boldsymbol{e}_+$ and $\boldsymbol{e}_-$ are both all-ones vectors of proper dimensions. For the sake of simplicity, we omit the bias terms b_+, b_- by $\boldsymbol{w}_+^\top = [\boldsymbol{u}_+^\top \ b_+]$ and $\boldsymbol{w}_-^\top = [\boldsymbol{u}_-^\top \ b_-]$. The two hyperplanes can be formulated as the following QPPs:

$$\begin{aligned} \min_{\boldsymbol{w}_+,\boldsymbol{\xi}} \ & \frac{1}{2}\|\boldsymbol{X}_+\boldsymbol{w}_+\|^2 + C_1\boldsymbol{e}_-^\top\boldsymbol{\xi} \\ s.t. \ & -\boldsymbol{X}_-\boldsymbol{w}_+ + \boldsymbol{\xi} \geq \boldsymbol{e}_-, \\ & \boldsymbol{\xi} \geq \boldsymbol{0}, \end{aligned} \tag{2}$$

and

$$\begin{aligned} \min_{\boldsymbol{w}_-,\boldsymbol{\eta}} \ & \frac{1}{2}\|\boldsymbol{X}_-\boldsymbol{w}_-\|^2 + C_2\boldsymbol{e}_+^\top\boldsymbol{\eta} \\ s.t. \ & \boldsymbol{X}_+\boldsymbol{w}_- + \boldsymbol{\eta} \geq \boldsymbol{e}_+, \\ & \boldsymbol{\eta} \geq \boldsymbol{0}, \end{aligned} \tag{3}$$

where C_1, C_2 are pre-specified penalty factors, $\boldsymbol{\xi}, \boldsymbol{\eta}$ are slack variables.

Once we obtain $\boldsymbol{w}_+$ and $\boldsymbol{w}_-$, the label of an arbitrary example $\boldsymbol{x}$ can be determined according to Eq. (4):

$$f(\boldsymbol{x}) = sgn\left(\left|\langle \boldsymbol{w}_-, \boldsymbol{x}\rangle\right| - \left|\langle \boldsymbol{w}_+, \boldsymbol{x}\rangle\right|\right). \tag{4}$$

2.2. S-TSVM

S-TSVM is a structural large margin classifier based on cluster granularity model (Xue, Chen, & Yang, 2011). It captures the structural information by adopting some clustering techniques.

Suppose we obtain two sets of clusters from the positive class (P) and the other from negative class (N). There are c_p clusters in P and c_n clusters in N. We define clusters $P_1, \ldots, P_{c_p}$ for class P and $N_1, \ldots, N_{c_n}$

for class N, respectively. The associated covariance matrices are denoted as $\hat{\boldsymbol{\Sigma}}_{P_i}$ and $\hat{\boldsymbol{\Sigma}}_{N_j}$ paired with the ith and jth clusters in class P and class N, $i = 1, \ldots, c_p, j = 1, \ldots, c_n$. Let $\boldsymbol{\Sigma}_{P_i} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{P_i} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix}$ and $\boldsymbol{\Sigma}_{N_i} = \begin{bmatrix} \hat{\boldsymbol{\Sigma}}_{N_i} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix}$, where $\mathbf{0}$ is a zero vector of proper dimensions. Similar to TSVM, S-TSVM also needs to construct two nonparallel hyperplanes given in Eq. (1). These two hyperplanes can be derived by addressing the following optimization problems respectively:

$$\begin{aligned} \min_{\boldsymbol{w}_+, \boldsymbol{\xi}} \quad & \frac{1}{2}\left\|\boldsymbol{X}_+\boldsymbol{w}_+\right\|^2 + \frac{1}{2}\left\|\boldsymbol{w}_+\right\|^2 + C_+\boldsymbol{e}_-^\top\boldsymbol{\xi} + \frac{1}{2}D_+\boldsymbol{w}_+^\top\boldsymbol{\Sigma}_+\boldsymbol{w}_+ \\ s.t. \quad & -\boldsymbol{X}_-\boldsymbol{w}_+ + \boldsymbol{\xi} \geq \boldsymbol{e}_-, \\ & \boldsymbol{\xi} \geq \mathbf{0}, \end{aligned} \tag{5}$$

and

$$\begin{aligned} \min_{\boldsymbol{w}_-, \boldsymbol{\eta}} \quad & \frac{1}{2}\left\|\boldsymbol{X}_-\boldsymbol{w}_-\right\|^2 + \frac{1}{2}\left\|\boldsymbol{w}_-\right\|^2 + C_-\boldsymbol{e}_+^\top\boldsymbol{\eta} + \frac{1}{2}D_-\boldsymbol{w}_-^\top\boldsymbol{\Sigma}_-\boldsymbol{w}_- \\ s.t. \quad & \boldsymbol{X}_+\boldsymbol{w}_- + \boldsymbol{\eta} \geq \boldsymbol{e}_+, \\ & \boldsymbol{\eta} \geq \mathbf{0}, \end{aligned} \tag{6}$$

where C_+, C_-, D_+, D_- are non-negative tradeoff parameters, $\boldsymbol{\xi}, \boldsymbol{\eta}$ are slack variables. By summing the covariance matrices of each cluster separately, the total covariance matrices for two classes can be derived as $\boldsymbol{\Sigma}_+ = \boldsymbol{\Sigma}_{P_1} + \cdots + \boldsymbol{\Sigma}_{P_{c_p}}$ and $\boldsymbol{\Sigma}_- = \boldsymbol{\Sigma}_{N_1} + \cdots + \boldsymbol{\Sigma}_{N_{c_n}}$. Additionally, $\boldsymbol{X}_+$, $\boldsymbol{X}_-$, $\boldsymbol{w}_+, \boldsymbol{w}_-$, $\boldsymbol{e}_+$ and $\boldsymbol{e}_-$ still follow the definition above. The predicted function of S-TSVM is as same as that of TSVM in Eq. (4).

2.3. MvTSVM

Consider a classification problem in views 1 and 2, MvTSVM aims to search for two nonparallel hyperplanes in each view: $f_+^{(1)} = \langle \boldsymbol{w}_+^{(1)}, \boldsymbol{x} \rangle$, $f_-^{(1)} = \langle \boldsymbol{w}_-^{(1)}, \boldsymbol{x} \rangle$ for view 1 and $f_+^{(2)} = \langle \boldsymbol{w}_+^{(2)}, \boldsymbol{x} \rangle$, $f_-^{(2)} = \langle \boldsymbol{w}_-^{(2)}, \boldsymbol{x} \rangle$ for view 2. For positive samples, MvTSVM keeps the consensus information by introducing a similarity constraint between two MvTSVMs. The same is true for negative samples. MvTSVM can be built as the following formulations:

$$\begin{aligned} \min_{\boldsymbol{w}_+^{(1)/(2)}, \boldsymbol{\xi}_-^{(1)/(2)}, \boldsymbol{\eta}_+} \quad & \frac{1}{2}\left\|\boldsymbol{X}_+^{(1)}\boldsymbol{w}_+^{(1)}\right\|^2 + \frac{1}{2}\left\|\boldsymbol{X}_+^{(2)}\boldsymbol{w}_+^{(2)}\right\|^2 + C_+^{(1)}\boldsymbol{e}_-^\top\boldsymbol{\xi}_-^{(1)} + C_+^{(2)}\boldsymbol{e}_-^\top\boldsymbol{\xi}_-^{(2)} \\ & + D_+\boldsymbol{e}_+^\top\boldsymbol{\eta}_+ \\ s.t. \quad & \left|\boldsymbol{X}_-^{(1)}\boldsymbol{w}_+^{(1)} - \boldsymbol{X}_-^{(2)}\boldsymbol{w}_+^{(2)}\right| \leq \boldsymbol{\eta}_+ + \epsilon_1, \\ & -\boldsymbol{X}_-^{(1)}\boldsymbol{w}_+^{(1)} \geq \boldsymbol{e}_- - \boldsymbol{\xi}_-^{(1)}, \\ & -\boldsymbol{X}_-^{(2)}\boldsymbol{w}_+^{(2)} \geq \boldsymbol{e}_- - \boldsymbol{\xi}_-^{(2)}, \\ & \boldsymbol{\xi}_-^{(1)} \geq \mathbf{0}, \boldsymbol{\xi}_-^{(2)} \geq \mathbf{0}, \boldsymbol{\eta}_+ \geq \mathbf{0}, \end{aligned} \tag{7}$$

and

$$\begin{aligned} \min_{\boldsymbol{w}_-^{(1)/(2)}, \boldsymbol{\xi}_+^{(1)/(2)}, \boldsymbol{\eta}_-} \quad & \frac{1}{2}\left\|\boldsymbol{X}_-^{(1)}\boldsymbol{w}_-^{(1)}\right\|^2 + \frac{1}{2}\left\|\boldsymbol{X}_-^{(2)}\boldsymbol{w}_-^{(2)}\right\|^2 + C_-^{(1)}\boldsymbol{e}_+^\top\boldsymbol{\xi}_+^{(1)} + C_-^{(2)}\boldsymbol{e}_+^\top\boldsymbol{\xi}_+^{(2)} \\ & + D_-\boldsymbol{e}_-^\top\boldsymbol{\eta}_- \\ s.t. \quad & \left|\boldsymbol{X}_+^{(1)}\boldsymbol{w}_-^{(1)} - \boldsymbol{X}_+^{(2)}\boldsymbol{w}_-^{(2)}\right| \leq \boldsymbol{\eta}_- + \epsilon_2, \\ & \boldsymbol{X}_+^{(1)}\boldsymbol{w}_-^{(1)} \geq \boldsymbol{e}_+ - \boldsymbol{\xi}_+^{(1)}, \\ & \boldsymbol{X}_+^{(2)}\boldsymbol{w}_-^{(2)} \geq \boldsymbol{e}_+ - \boldsymbol{\xi}_+^{(2)}, \\ & \boldsymbol{\xi}_+^{(1)} \geq \mathbf{0}, \boldsymbol{\xi}_+^{(2)} \geq \mathbf{0}, \boldsymbol{\eta}_- \geq \mathbf{0}, \end{aligned} \tag{8}$$

where $\boldsymbol{X}_+^{(1)} = [\boldsymbol{A}_1\ \boldsymbol{e}_+]$, $\boldsymbol{X}_+^{(2)} = [\boldsymbol{A}_2\ \boldsymbol{e}_+]$, $\boldsymbol{X}_-^{(1)} = [\boldsymbol{B}_1\ \boldsymbol{e}_-]$, $\boldsymbol{X}_-^{(2)} = [\boldsymbol{B}_2\ \boldsymbol{e}_-]$. Matrices $\boldsymbol{A}_1, \boldsymbol{A}_2$ symbolize the positive datasets in views 1 and 2, and $\boldsymbol{B}_1, \boldsymbol{B}_2$ symbolize the negative datasets for two views respectively. $\boldsymbol{\xi}_+^{(1)/(2)}$, $\boldsymbol{\xi}_-^{(1)/(2)}$ are slack variables. $C_+^{(1)}, C_+^{(2)}, C_-^{(1)}, C_-^{(2)}, D_+, D_-, \epsilon_1, \epsilon_2$ are parameters chosen in advance.

After solving the dual problem corresponding to the primal problem, the function of four hyperplanes can be obtained. For a test sample $\boldsymbol{x} = \left(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}\right)$, its label y depends on the difference as

$$g(\boldsymbol{x}) = \left|f_+^{(1)} + f_+^{(2)}\right| - \left|f_-^{(1)} + f_-^{(2)}\right|. \tag{9}$$

If $g(\boldsymbol{x}) > 0$, y is equal to -1, $\boldsymbol{x}$ will be judged as negative, otherwise positive.

3. Multi-view structural twin support vector machine (MvSTSVM)

Since S-TSVM exploits the prior knowledge based on structural information, it is more efficient for classification compared to traditional SVM and TSVM. In the following section, we extend S-TSVM to a new multi-view structural classifier—MvSTSVM. Our research includes three parts: (1) structural information extraction within classes; (2) linear and nonlinear cases of our proposed MvSTSVM; (3) comparison with other multi-view models. The key notations are defined in Table 1.

3.1. Clustering

Traditional SVM classifiers pay more attention to separating different kinds of samples and ignore the structural information among the training data within classes. However, SRSVM shows that these classifiers are sensitive to the distribution of data which introduces the prior knowledge for our classifiers (Xue et al., 2011). As for MVL, MvSLMC adds this information into multi-view SVM based on cluster granularity model (Yeung, Wang, Ng, Tsang, & Wang, 2007; Zhao & Xu, 2023).

Following the previous research, clustering is utilized in MvSTSVM to capture the data distribution among classes and views. Some typical clustering methods can be employed in this step such as k-means (Hartigan & Wong, 1979), mean-shift (Derpanis, 2005) and DBSCAN (Ester, Kriegel, Sander, Xu, et al., 1996). Similar to MvSLMC, we employ Ward's linkage clustering (Ward Jr, 1963), a type of hierarchical clustering method. This method is convenient for us to calculate the covariance matrix and (weighted) Mahalanobis distances since it can derive compact and spherical clusters.

At first, each sample is treated as a cluster to participate in clustering. The Ward's linkage between two examples $\boldsymbol{x}_i$, $\boldsymbol{x}_j$ is

$$W(\boldsymbol{x}_i, \boldsymbol{x}_j) = \frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{2}. \tag{10}$$

Furthermore, if cluster P and cluster N are given, their Ward's linkage $W(P, N)$ can be constructed as

$$W(P, N) = \frac{|P| \cdot |N| \cdot \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_N\|^2}{(|P| + |N|)}, \tag{11}$$

where $\boldsymbol{\mu}_P$ and $\boldsymbol{\mu}_N$ stand for the mean value of samples in P and N, respectively. $|P|$ and $|N|$ denote the number of samples in P and N. In the process of clustering, the two closest clusters are grouped first. Thus, the distance between clusters increases as clusters merge. We select the number "0" in Handwritten digits[1] and depict this process on the merge distance curve, as shown in Fig. 1.

It is a key step to decide the number of clusters before clustering because the total number of clusters affects the clustering and classification results indirectly. Salvador and Chan (Salvador & Chan, 2004) propose that the number of clusters corresponds to the point of the largest curvature on the merge distance curve. This point is named "knee point". In Fig. 1, the red dot is the knee point.

In MVL, clustering techniques will be adopted in each class from each view. That means a sample may belong to cluster P in view 1, but belong to cluster N in view 2. Different from MvSLMC, MvSTSVM only cares about the structural information of one class in each model rather than considering positive and negative classes simultaneously. In other

[1] http://archive.ics.uci.edu/ml/datasets/Multiple+Features

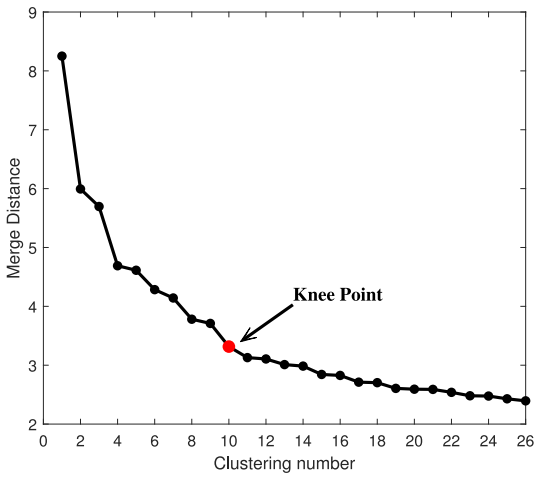


Fig. 1. Determining the number of clusters.

Table 1
List of notations.

Notations	Definition
$c_{\pm}^{(v)}$	total number of positive/negative clusters in view v
$\boldsymbol{x}_{\pm,k}^{(v)}$	the kth positive/negative training sample in view v
$\boldsymbol{P}_c^{(v)}/\boldsymbol{N}_c^{(v)}$	the cth positive/negative cluster of the vth view
$\lvert\boldsymbol{P}_c^{(v)}\rvert/\lvert\boldsymbol{N}_c^{(v)}\rvert$	the size of samples in cluster $\boldsymbol{P}_c^{(v)}/\boldsymbol{N}_c^{(v)}$
$\boldsymbol{\mu}_{\boldsymbol{P}_c^{(v)}}^{(z)}/\boldsymbol{\mu}_{\boldsymbol{N}_c^{(v)}}^{(z)}$	mean of dataset $\{\boldsymbol{x}_{+,k}^{(z)}\lvert k\in\boldsymbol{P}_c^{(v)}\}/\{\boldsymbol{x}_{-,k}^{(z)}\lvert k\in\boldsymbol{N}_c^{(v)}\}$
$\boldsymbol{u}_{\pm}^{(v)}$	weight vectors of positive/negative hyperplane in view v
$\theta_{\pm}^{(v)}$	view weights of positive/negative hyperplane in view v
p	the total dimension of training set
p_v	the dimension of training data in view v
$\boldsymbol{C}_v$	the training samples of the vth view
$\boldsymbol{A}_v/\boldsymbol{B}_v$	positive/negative training set of the vth view
n_1/n_2	the size of positive/negative training samples
$\lvert\boldsymbol{a}\rvert$, $\lVert\boldsymbol{a}\rVert$	1-norm and 2-norm of vector $\boldsymbol{a}$
$\langle\boldsymbol{a},\boldsymbol{b}\rangle$	the inner product of vector $\boldsymbol{a}$ and vector $\boldsymbol{b}$

words, the structural information of the positive class is independent and unrelated to the negative one during the training process. Suppose we have clustered the positive training set in each view.

Afterward, the structural information of these clusters can be dug out by computing their covariance matrices. The covariance matrices of cluster $P_c^{(v)}$ from the zth view is

$$\boldsymbol{\Sigma}_{P_c^{(v)}}^{(z)}=\frac{1}{|P_c^{(v)}|}\sum_{k\in P_c^{(v)}}\boldsymbol{S}\boldsymbol{S}^{\top},\tag{12}$$

where $\boldsymbol{S}=\boldsymbol{x}_{+,k}^{(z)}-\boldsymbol{\mu}_{P_c^{(v)}}^{(z)}$. Meanwhile, $|P_c^{(v)}|,\boldsymbol{x}_{+,k}^{(z)},P_c^{(v)},\boldsymbol{\mu}_{P_c^{(v)}}^{(z)}$ are defined in Table 1.

Furthermore, we can calculate the covariance of cluster $P_c^{(v)}$ in any other views as long as obtaining the index of all positive training samples. At this time, structural information can transfer from one view to another view.

In view u, the covariance matrix of distinct clusters can be added up directly to derive the total covariance matrix $\hat{\boldsymbol{\Sigma}}_{+}^{(z)}$ in the positive class. But in fact, each view is not equally important to the classifier. Therefore, we combine view weights with the total covariance matrix in positive class. Hence, $\hat{\boldsymbol{\Sigma}}_{+}^{(z)}$ can be obtained as

$$\hat{\boldsymbol{\Sigma}}_{+}^{(z)}=\sum_{v=1}^{V}\theta_{+}^{(v)}\sum_{c=1}^{c_{+}^{(v)}}\boldsymbol{\Sigma}_{P_c^{(v)}}^{(z)},\tag{13}$$

where V denotes the total number of views, $\theta_{+}^{(v)}$ is the weight in view v for positive class. In the same way, we can also get the total covariance matrix of the negative class through Eq. (12):

$$\hat{\boldsymbol{\Sigma}}_{-}^{(z)}=\sum_{v=1}^{V}\theta_{-}^{(v)}\sum_{c=1}^{c_{-}^{(v)}}\boldsymbol{\Sigma}_{N_c^{(v)}}^{(z)},\tag{14}$$

where $\theta_{-}^{(v)}$ is the weight in view v for negative class. Certainly, $\hat{\boldsymbol{\Sigma}}_{+}^{(z)}$ and $\hat{\boldsymbol{\Sigma}}_{-}^{(z)}$ are both symmetric and semi-positive definite matrices.

3.2. The MvSTSVM model

3.2.1. Linear MvSTSVM

For a V-view binary classification problem, positive and negative datasets are given. Given an arbitrary sample $\boldsymbol{x}$, it is written as $\boldsymbol{x}=(\boldsymbol{x}^{(1)^{\top}},\ldots,\boldsymbol{x}^{(V)^{\top}})^{\top}$. Suppose that the positive dataset of vth view can be expressed as matrix $\boldsymbol{A}_v$. $\boldsymbol{x}_{+,i}^{(v)}$ is the ith row of $\boldsymbol{A}_v$, representing the ith positive sample from view v. Similarly, matrix $\boldsymbol{B}_v$ stands for the negative dataset of the vth view. $\boldsymbol{x}_{-,j}^{(v)}$ denotes the jth negative sample of the vth view. MvSTSVM intends to find two nonparallel hyperplanes for each view:

$$f_{+}^{(v)}:\langle\boldsymbol{u}_{+}^{(v)},\boldsymbol{x}^{(v)}\rangle+b_{+}^{(v)}=0\quad\text{and}\quad f_{-}^{(v)}:\langle\boldsymbol{u}_{-}^{(v)},\boldsymbol{x}^{(v)}\rangle+b_{-}^{(v)}=0.\tag{15}$$

For the sake of simplicity, we define matrices $\boldsymbol{X}_{+}^{(v)},\boldsymbol{X}_{-}^{(v)}$ and vectors $\boldsymbol{w}_{+}^{(v)},\boldsymbol{w}_{-}^{(v)}$ as

$$\boldsymbol{X}_{+}^{(v)}=[\boldsymbol{A}_v\ \boldsymbol{e}_{+}],\ \boldsymbol{w}_{+}^{(v)^{\top}}=[\boldsymbol{u}_{+}^{(v)^{\top}}\ b_{+}^{(v)}],$$
$$\boldsymbol{X}_{-}^{(v)}=[\boldsymbol{B}_v\ \boldsymbol{e}_{-}],\ \boldsymbol{w}_{-}^{(v)^{\top}}=[\boldsymbol{u}_{-}^{(v)^{\top}}\ b_{-}^{(v)}],$$

where $v=1,\ldots,V$, $\boldsymbol{e}_{+}\in\mathbb{R}^{n_1},\boldsymbol{e}_{-}\in\mathbb{R}^{n_2}$.

Fig. 2 shows the construction of MvSTSVM. Our model introduces the view perspective into multi-view data to dig the information between different perspectives. Meanwhile, MvSTSVM collects prior structural information within the class by clustering methods. Therefore, MvSTSVM satisfies both consensus and complementarity principles.

The optimization models for linear MvSTSVM can be formulated as

$$\begin{aligned}\min_{\boldsymbol{w}_{+}^{(v)},\boldsymbol{\theta}_{+},\boldsymbol{\xi}_{+}^{(v)}}\ &\frac{1}{2}\sum_{v=1}^{V}\theta_{+}^{(v)}\left\|\boldsymbol{X}_{+}^{(v)}\boldsymbol{w}_{+}^{(v)}\right\|^2+\frac{1}{2}\sum_{v=1}^{V}\theta_{+}^{(v)}\left\|\boldsymbol{w}_{+}^{(v)}\right\|^2+\frac{\lambda}{2}\|\boldsymbol{\theta}_{+}\|^2\\&+C_{+}\sum_{v=1}^{V}\boldsymbol{e}_{-}^{\top}\boldsymbol{\xi}_{+}^{(v)}+\frac{D_{+}}{2}\sum_{v=1}^{V}\boldsymbol{w}_{+}^{(v)^{\top}}\boldsymbol{\Sigma}_{+}^{(v)}\boldsymbol{w}_{+}^{(v)}\\ s.t.\ \ &-\boldsymbol{X}_{-}^{(v)}\boldsymbol{w}_{+}^{(v)}\geq\boldsymbol{e}_{-}-\boldsymbol{\xi}_{+}^{(v)},\boldsymbol{\xi}_{+}^{(v)}\geq\boldsymbol{0},\\&\sum_{v=1}^{V}\theta_{+}^{(v)}=1,\theta_{+}^{(v)}\geq0,v=1,\ldots,V,\end{aligned}\tag{16}$$

and

$$\begin{aligned}\min_{\boldsymbol{w}_{-}^{(v)},\boldsymbol{\theta}_{-},\boldsymbol{\xi}_{-}^{(v)}}\ &\frac{1}{2}\sum_{v=1}^{V}\theta_{-}^{(v)}\left\|\boldsymbol{X}_{-}^{(v)}\boldsymbol{w}_{-}^{(v)}\right\|^2+\frac{1}{2}\sum_{v=1}^{V}\theta_{-}^{(v)}\left\|\boldsymbol{w}_{-}^{(v)}\right\|^2+\frac{\eta}{2}\|\boldsymbol{\theta}_{-}\|^2\\&+C_{-}\sum_{v=1}^{V}\boldsymbol{e}_{+}^{\top}\boldsymbol{\xi}_{-}^{(v)}+\frac{D_{-}}{2}\sum_{v=1}^{V}\boldsymbol{w}_{-}^{(v)^{\top}}\boldsymbol{\Sigma}_{-}^{(v)}\boldsymbol{w}_{-}^{(v)}\\ \text{s.t.}\ \ &\boldsymbol{X}_{+}^{(v)}\boldsymbol{w}_{-}^{(v)}\geq\boldsymbol{e}_{+}-\boldsymbol{\xi}_{-}^{(v)},\boldsymbol{\xi}_{-}^{(v)}\geq\boldsymbol{0},\\&\sum_{v=1}^{V}\theta_{-}^{(v)}=1,\theta_{-}^{(v)}\geq0,v=1,\ldots,V,\end{aligned}\tag{17}$$

where $\boldsymbol{\Sigma}_{+}^{(v)}=\begin{bmatrix}\hat{\boldsymbol{\Sigma}}_{+}^{(v)}&\boldsymbol{0}\\\boldsymbol{0}&0\end{bmatrix}$ and $\boldsymbol{\Sigma}_{-}^{(v)}=\begin{bmatrix}\hat{\boldsymbol{\Sigma}}_{-}^{(v)}&\boldsymbol{0}\\\boldsymbol{0}&0\end{bmatrix}$, $\boldsymbol{0}$ is a zero vector of proper dimensions. $\boldsymbol{e}_{+}\in\mathbb{R}^{n_1},\boldsymbol{e}_{-}\in\mathbb{R}^{n_2}$ are all-ones vectors of appropriate dimensions. $C_{+},C_{-},D_{+},D_{-},\lambda,\eta$ are predefined tradeoff factors and $\boldsymbol{\xi}_{+}^{(v)},\boldsymbol{\xi}_{-}^{(v)}$ are slack variables for two hyperplanes in the vth view.

Furthermore, we take problem (16) as an example to propose the following explanation for MvSTSVM.

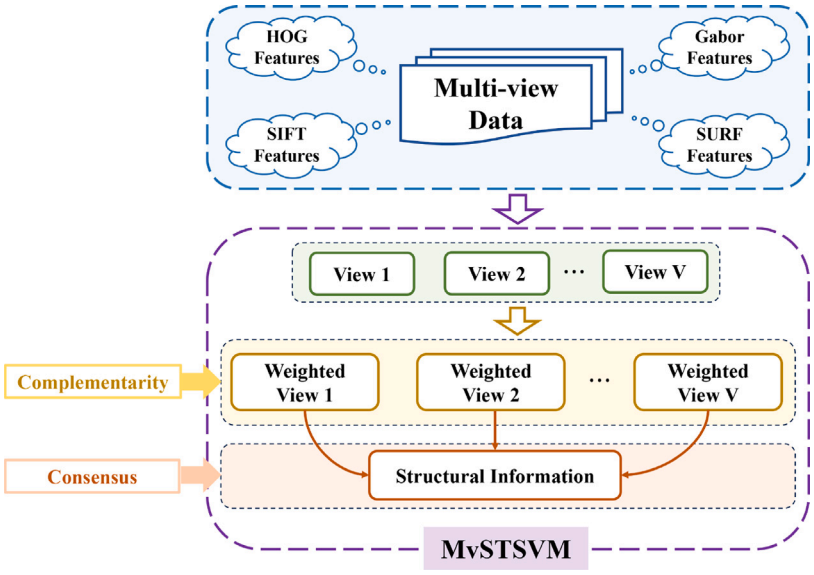


Fig. 2. Basic composition of MvSTSVM.

- $\theta_+^{(v)}$ stands for the positive view weight of the vth view so that the complementarity principle among views can be guaranteed. Different from most of the multi-view models, the view weights of MvSTSVM in two optimization problems are independent rather than shared, which makes the solution of our model more flexible. Besides, $\frac{\lambda}{2}\|\boldsymbol{\theta}_+\|^2$ can avoid the trivial solution and make problem (31) become the classical QPP, simultaneously.
- $\frac{1}{2}\sum_{v=1}^{V}\theta_+^{(v)}\left\|\boldsymbol{X}_+^{(v)}\boldsymbol{w}_+^{(v)}\right\|^2$ is designed to make the positive hyperplanes of distinct views closer to the positive samples. In addition, the regularization item $\frac{1}{2}\sum_{v=1}^{V}\theta_+^{(v)}\left\|\boldsymbol{w}_+^{(v)}\right\|^2$ maximizes the soft margin and ensures two classes separated sufficiently. These two items enable MvSTSVM to achieve both intra-class compactness and inter-class separability.
- $\boldsymbol{\Sigma}_+^{(v)}$ is the total covariance matrix of the vth perspective in positive class, following the definition in Eq. (13). Besides, $\frac{D_+}{2}\sum_{v=1}^{V}\boldsymbol{w}_+^{(v)\top}\boldsymbol{\Sigma}_+^{(v)}\boldsymbol{w}_+^{(v)}$ introduces the structural information of other views into the current view and enhances the compactness within positive classes.

In models (16) and (17), both $\{\boldsymbol{w}_\pm^{(v)}\}_{v=1}^V$ and $\{\theta_\pm^{(v)}\}_{v=1}^V$ are unknown variables that need to be optimized. It will be more complicated if we solve the optimization model directly. Accordingly, we adopt an alternating strategy to iteratively update for the approximate optimal solution. This iterative strategy mainly includes the following two steps:

Fix $\{\theta_\pm^{(v)}\}_{v=1}^V$ **and update** $\{\boldsymbol{w}_\pm^{(v)}\}_{v=1}^V$: When $\{\theta_\pm^{(v)}\}_{v=1}^V$ is given, optimization problems (16) and (17) can be simplified problems (18) and (19) to obtain the approximate solution $\{\boldsymbol{w}_\pm^{(v)}\}_{v=1}^V$.

$$\begin{aligned}\min_{\{\boldsymbol{w}_+^{(v)}\}_{v=1}^V}\quad & \frac{1}{2}\sum_{v=1}^{V}\theta_+^{(v)}\left\|\boldsymbol{X}_+^{(v)}\boldsymbol{w}_+^{(v)}\right\|^2+\frac{1}{2}\sum_{v=1}^{V}\theta_+^{(v)}\left\|\boldsymbol{w}_+^{(v)}\right\|^2+C_+\sum_{v=1}^{V}\boldsymbol{e}_-^\top\boldsymbol{\xi}_+^{(v)}\\ &+\frac{D_+}{2}\sum_{v=1}^{V}\boldsymbol{w}_+^{(v)\top}\boldsymbol{\Sigma}_+^{(v)}\boldsymbol{w}_+^{(v)}\\ s.t.\quad & -\boldsymbol{X}_-^{(v)}\boldsymbol{w}_+^{(v)}\geq\boldsymbol{e}_--\boldsymbol{\xi}_+^{(v)},\boldsymbol{\xi}_+^{(v)}\geq\boldsymbol{0},v=1,\ldots,V,\end{aligned}\tag{18}$$

and

$$\begin{aligned}\min_{\{\boldsymbol{w}_-^{(v)}\}_{v=1}^V}\quad & \frac{1}{2}\sum_{v=1}^{V}\theta_-^{(v)}\left\|\boldsymbol{X}_-^{(v)}\boldsymbol{w}_-^{(v)}\right\|^2+\frac{1}{2}\sum_{v=1}^{V}\theta_-^{(v)}\left\|\boldsymbol{w}_-^{(v)}\right\|^2+C_-\sum_{v=1}^{V}\boldsymbol{e}_+^\top\boldsymbol{\xi}_-^{(v)}\\ &+\frac{D_-}{2}\sum_{v=1}^{V}\boldsymbol{w}_-^{(v)\top}\boldsymbol{\Sigma}_-^{(v)}\boldsymbol{w}_-^{(v)}\\ s.t.\quad & \boldsymbol{X}_+^{(v)}\boldsymbol{w}_-^{(v)}\geq\boldsymbol{e}_+-\boldsymbol{\xi}_-^{(v)},\boldsymbol{\xi}_-^{(v)}\geq\boldsymbol{0},v=1,\ldots,V.\end{aligned}\tag{19}$$

The Lagrangian function of problem (18) is constructed as

$$\begin{aligned}L=&\frac{1}{2}\sum_{v=1}^{V}\theta_+^{(v)}\left\|\boldsymbol{X}_+^{(v)}\boldsymbol{w}_+^{(v)}\right\|^2+\frac{1}{2}\sum_{v=1}^{V}\theta_+^{(v)}\left\|\boldsymbol{w}_+^{(v)}\right\|^2+C_+\sum_{v=1}^{V}\boldsymbol{e}_-^\top\boldsymbol{\xi}_+^{(v)}\\ &+\frac{D_+}{2}\sum_{v=1}^{V}\boldsymbol{w}_+^{(v)\top}\boldsymbol{\Sigma}_+^{(v)}\boldsymbol{w}_+^{(v)}-\sum_{v=1}^{V}\boldsymbol{\alpha}_+^{(v)\top}(-\boldsymbol{X}_-^{(v)}\boldsymbol{w}_+^{(v)}-\boldsymbol{e}_-+\boldsymbol{\xi}_+^{(v)})\\ &-\sum_{v=1}^{V}\boldsymbol{\beta}_+^{(v)\top}\boldsymbol{\xi}_+^{(v)},\end{aligned}\tag{20}$$

where $\boldsymbol{\alpha}_+^{(v)}$ and $\boldsymbol{\beta}_+^{(v)}$ are Lagrange multiplier vectors. Then we receive the following Karush–Kuhn–Tucker (KKT) conditions through differentiating L to variables $\boldsymbol{w}_+^{(v)}$ and $\boldsymbol{\xi}_+^{(v)}$:

$$\frac{\partial L}{\partial\boldsymbol{w}_+^{(v)}}=\theta_+^{(v)}\boldsymbol{X}_+^{(v)\top}\boldsymbol{X}_+^{(v)}\boldsymbol{w}_+^{(v)}+\theta_+^{(v)}\boldsymbol{w}_+^{(v)}+\boldsymbol{D}_+\boldsymbol{\Sigma}_+^{(v)}\boldsymbol{w}_+^{(v)}+\boldsymbol{X}_-^{(v)\top}\boldsymbol{\alpha}_+^{(v)}=\boldsymbol{0},\tag{21}$$

$$\frac{\partial L}{\partial\boldsymbol{\xi}_+^{(v)}}=C_+\boldsymbol{e}_--\boldsymbol{\beta}_+^{(v)}-\boldsymbol{\alpha}_+^{(v)}=\boldsymbol{0},\tag{22}$$

$$\alpha_{+,i}^{(v)}(-\boldsymbol{w}_+^{(v)\top}\boldsymbol{x}_{-,i}^{(v)}-1+\xi_{+,i}^{(v)})=0,\tag{23}$$

$$\beta_{+,i}^{(v)}\xi_{+,i}^{(v)}=0.\tag{24}$$

According to Eq. (21), the solution of $\boldsymbol{w}_+^{(v)}$ can be derived in the following way:

$$\boldsymbol{w}_+^{(v)}=-\left(\theta_+^{(v)}\boldsymbol{X}_+^{(v)\top}\boldsymbol{X}_+^{(v)}+\theta_+^{(v)}\boldsymbol{I}_++D_+\boldsymbol{\Sigma}_+^{(v)}\right)^{-1}\boldsymbol{X}_-^{(v)\top}\boldsymbol{\alpha}_+^{(v)},\tag{25}$$

where $\boldsymbol{I}_+$ represents the identity matrix with identical dimension as $\boldsymbol{\Sigma}_+^{(v)}$. Likewise, we can write the Lagrangian function of the problem (19) and get

$$\boldsymbol{w}_-^{(v)}=\left(\theta_-^{(v)}\boldsymbol{X}_-^{(v)\top}\boldsymbol{X}_-^{(v)}+\theta_-^{(v)}\boldsymbol{I}_-+D_-\boldsymbol{\Sigma}_-^{(v)}\right)^{-1}\boldsymbol{X}_+^{(v)\top}\boldsymbol{\alpha}_-^{(v)},\tag{26}$$

where $\boldsymbol{I}_-$ represents the identity matrix with identical dimension as $\boldsymbol{\Sigma}_-^{(v)}$.

Finally, dual formulations of problems (18) and (19) are generated as follows:

$$\begin{aligned}\min_{\{\boldsymbol{\alpha}_+^{(v)}\}_{v=1}^V}\quad & \sum_{v=1}^{V}\left(\frac{1}{2}\boldsymbol{\alpha}_+^{(v)\top}\boldsymbol{R}_+^{(v)}\boldsymbol{\alpha}_+^{(v)}-\boldsymbol{e}_-^\top\boldsymbol{\alpha}_+^{(v)}\right)\\ s.t.\quad & \boldsymbol{0}\leq\boldsymbol{\alpha}_+^{(v)}\leq C_+\boldsymbol{e}_-,v=1,\ldots,V,\end{aligned}\tag{27}$$

and

$$\min_{\{\boldsymbol{\alpha}_-^{(v)}\}_{v=1}^V} \sum_{v=1}^{V} \left(\frac{1}{2} \boldsymbol{\alpha}_-^{(v)\top} \boldsymbol{R}_-^{(v)} \boldsymbol{\alpha}_-^{(v)} - \boldsymbol{e}_+^\top \boldsymbol{\alpha}_-^{(v)} \right)$$
$$s.t. \quad \mathbf{0} \le \boldsymbol{\alpha}_-^{(v)} \le C_- \boldsymbol{e}_+, v = 1, \dots, V, \tag{28}$$

where

$$\boldsymbol{R}_+^{(v)} = \boldsymbol{X}_-^{(v)} (\theta_+^{(v)} \boldsymbol{X}_+^{(v)\top} \boldsymbol{X}_+^{(v)} + \theta_+^{(v)} \boldsymbol{I}_+ + D_+ \boldsymbol{\Sigma}_+^{(v)})^{-1} \boldsymbol{X}_-^{(v)\top}, \tag{29}$$
$$\boldsymbol{R}_-^{(v)} = \boldsymbol{X}_+^{(v)} (\theta_-^{(v)} \boldsymbol{X}_-^{(v)\top} \boldsymbol{X}_-^{(v)} + \theta_-^{(v)} \boldsymbol{I}_- + D_- \boldsymbol{\Sigma}_-^{(v)})^{-1} \boldsymbol{X}_+^{(v)\top}, \tag{30}$$

$\boldsymbol{\alpha}_+^{(v)}$, $\boldsymbol{\alpha}_-^{(v)}$ are Lagrange multiplier vectors, $\boldsymbol{I}_+, \boldsymbol{I}_-$ are identity matrices of proper dimensions. To deal with the dual formulation more conveniently, every dual formulation can be divided into V small-scale problems for solution.

Fix $\{\boldsymbol{w}_\pm^{(v)}\}_{v=1}^V$ and update $\{\theta_\pm^{(v)}\}_{v=1}^V$: When $\{\boldsymbol{w}_\pm^{(v)}\}_{v=1}^V$ is obtained by problems (18) and (19), the optimization problems (16) and (17) can be simplified as problems (31) and (32) to get the approximate solution $\{\theta_\pm^{(v)}\}_{v=1}^V$.

$$\min_{\theta_+} \quad \frac{1}{2} \boldsymbol{F}^\top \theta_+ + \frac{1}{2} \boldsymbol{G}^\top \theta_+ + \frac{D_+}{2} \boldsymbol{Q}^\top \theta_+ + \frac{\lambda}{2} \|\theta_+\|^2$$
$$s.t. \quad \theta_+^\top \boldsymbol{e} = 1, \tag{31}$$

and

$$\min_{\theta_-} \quad \frac{1}{2} \boldsymbol{J}^\top \theta_- + \frac{1}{2} \boldsymbol{K}^\top \theta_- + \frac{D_-}{2} \boldsymbol{P}^\top \theta_- + \frac{\eta}{2} \|\theta_-\|^2$$
$$s.t. \quad \theta_-^\top \boldsymbol{e} = 1, \tag{32}$$

where

$$\boldsymbol{F} = (\|\boldsymbol{X}_+^{(1)} \boldsymbol{w}_+^{(1)}\|^2, \|\boldsymbol{X}_+^{(2)} \boldsymbol{w}_+^{(2)}\|^2, \dots, \|\boldsymbol{X}_+^{(V)} \boldsymbol{w}_+^{(V)}\|^2)^\top, \tag{33}$$
$$\boldsymbol{J} = (\|\boldsymbol{X}_-^{(1)} \boldsymbol{w}_-^{(1)}\|^2, \|\boldsymbol{X}_-^{(2)} \boldsymbol{w}_-^{(2)}\|^2, \dots, \|\boldsymbol{X}_-^{(V)} \boldsymbol{w}_-^{(V)}\|^2)^\top, \tag{34}$$
$$\boldsymbol{G} = (\|\boldsymbol{w}_+^{(1)}\|^2, \|\boldsymbol{w}_+^{(2)}\|^2, \dots, \|\boldsymbol{w}_+^{(V)}\|^2)^\top, \tag{35}$$
$$\boldsymbol{K} = (\|\boldsymbol{w}_-^{(1)}\|^2, \|\boldsymbol{w}_-^{(2)}\|^2, \dots, \|\boldsymbol{w}_-^{(V)}\|^2)^\top, \tag{36}$$
$$\boldsymbol{Q} = (\sum_{v=1}^{V} \sum_{c=1}^{c_+^{(1)}} \boldsymbol{w}_+^{(v)\top} \boldsymbol{\Sigma}_{P_c^{(1)}}^{(v)} \boldsymbol{w}_+^{(v)}, \sum_{v=1}^{V} \sum_{c=1}^{c_+^{(2)}} \boldsymbol{w}_+^{(v)\top} \boldsymbol{\Sigma}_{P_c^{(2)}}^{(v)} \boldsymbol{w}_+^{(v)}, \dots, \sum_{v=1}^{V} \sum_{c=1}^{c_+^{(V)}} \boldsymbol{w}_+^{(v)\top} \boldsymbol{\Sigma}_{P_c^{(V)}}^{(v)} \boldsymbol{w}_+^{(v)})^\top, \tag{37}$$
$$\boldsymbol{P} = (\sum_{v=1}^{V} \sum_{c=1}^{c_-^{(1)}} \boldsymbol{w}_-^{(v)\top} \boldsymbol{\Sigma}_{N_c^{(1)}}^{(v)} \boldsymbol{w}_-^{(v)}, \sum_{v=1}^{V} \sum_{c=1}^{c_-^{(2)}} \boldsymbol{w}_-^{(v)\top} \boldsymbol{\Sigma}_{N_c^{(2)}}^{(v)} \boldsymbol{w}_-^{(v)}, \dots, \sum_{v=1}^{V} \sum_{c=1}^{c_-^{(V)}} \boldsymbol{w}_-^{(v)\top} \boldsymbol{\Sigma}_{N_c^{(V)}}^{(v)} \boldsymbol{w}_-^{(v)})^\top, \tag{38}$$
$$\theta_+ = (\theta_+^{(1)}, \theta_+^{(2)}, \dots, \theta_+^{(V)})^\top, \theta_- = (\theta_-^{(1)}, \theta_-^{(2)}, \dots, \theta_-^{(V)})^\top. \tag{39}$$

Algorithm 1 Linear MvSTSVM

1: **Input**: training set $\mathcal{X} = \{(\boldsymbol{x}_i^{(1)}, \dots, \boldsymbol{x}_i^{(V)}, y_i)\}_{i=1}^n$, parameters $(C_+, C_-, D_+, D_-, \lambda, \eta)$, maximum iterations t_{max};
2: **Initialize**: $\theta_+^0 = \frac{1}{V}, \theta_-^0 = \frac{1}{V}, t = t' = 0$;
3: **While** $\|\theta_+^{t+1} - \theta_+^t\| > \delta$ and $t \le t_{max}$ **do**
4: acquire $\boldsymbol{w}_+^t$ by solving dual problem (27);
5: acquire θ_+^{t+1} according to the optimization model (31);
6: $t = t + 1$;
7: **End While**
8: **While** $\|\theta_-^{t'+1} - \theta_-^{t'}\| > \delta$ and $t' \le t_{max}$ **do**
9: acquire $\boldsymbol{w}_-^{t'}$ by solving dual problem (28);
10: acquire $\theta_-^{t'+1}$ according to the optimization model (32);
11: $t' = t' + 1$;
12: **End While**
13: **Output**: given a test sample $\boldsymbol{x}$, its label y depends on predictor (40).

Once we turn up the best solution $\{\boldsymbol{w}_\pm^{(v)*}\}_{v=1}^V$ and $\{\theta_\pm^{(v)*}\}_{v=1}^V$ from problems (27)–(28) and (31)–(32), the predicted function of MvSTSVM can be obtained. For a multi-view test sample $\boldsymbol{x}$, its label is determined by the predicted function

$$f(\boldsymbol{x}) = \sum_{v=1}^{V} \theta_+^{(v)*} |[\boldsymbol{x}^{(v)\top} \ 1] \boldsymbol{w}_+^{(v)*}| - \sum_{v=1}^{V} \theta_-^{(v)*} |[\boldsymbol{x}^{(v)\top} \ 1] \boldsymbol{w}_-^{(v)*}|. \tag{40}$$

If $f(\boldsymbol{x}) > 0$, $\boldsymbol{x}$ is classified as negative class, otherwise positive class. The algorithm of linear MvSTSVM is concluded in Algorithm 1.

3.2.2. Nonlinear MvSTSVM

When the training samples are linearly indivisible, the performance of linear classifiers will be poor. Fortunately, this situation can be improved if the samples from a low-dimensional space are mapped to a space with a higher dimension. This method leads to the separation of linearly inseparable samples. In this part, we extend the linear MvSTSVM to a nonlinear version.

Suppose training sample $\boldsymbol{x}$ is mapped to a nonlinear high-dimensional Hilbert space $\mathcal{H}$. In this case, $\boldsymbol{u}_+^{(v)}$ and $\boldsymbol{u}_-^{(v)}$ are denoted as $\boldsymbol{u}_+^{(v)} = \boldsymbol{k}_v \boldsymbol{\rho}_+^{(v)}$ and $\boldsymbol{u}_-^{(v)} = \boldsymbol{k}_v \boldsymbol{\rho}_-^{(v)}$ (Qi et al., 2013), where $\boldsymbol{k}_v = \Phi(\boldsymbol{C}_v)$ and $\boldsymbol{C}_v = [\boldsymbol{A}_v^\top, \boldsymbol{B}_v^\top]^\top$. Then the two kernel-generated hyperplanes can be built as:

$$K(\boldsymbol{x}^{(v)\top}, \boldsymbol{C}_v^\top) \boldsymbol{\rho}_+^{(v)} + b_+^{(v)} = 0 \quad textand \quad K(\boldsymbol{x}^{(v)\top}, \boldsymbol{C}_v^\top) \boldsymbol{\rho}_-^{(v)} + b_-^{(v)} = 0. \tag{41}$$

Additionally, we define

$$\boldsymbol{X}_+^{(v)} = [K(\boldsymbol{A}_v, \boldsymbol{C}_v^\top) \ \boldsymbol{e}_+], \boldsymbol{w}_+^{(v)\top} = [\boldsymbol{\rho}_+^{(v)\top} \ b_+^{(v)}],$$
$$\boldsymbol{X}_-^{(v)} = [K(\boldsymbol{B}_v, \boldsymbol{C}_v^\top) \ \boldsymbol{e}_-], \boldsymbol{w}_-^{(v)\top} = [\boldsymbol{\rho}_-^{(v)\top} \ b_-^{(v)}],$$

where $\boldsymbol{e}_+, \boldsymbol{e}_-$ are vectors of ones. K is a chosen kernel function and $K(\boldsymbol{x}_i, \boldsymbol{x}_j) = (\Phi(\boldsymbol{x}_i) \cdot \Phi(\boldsymbol{x}_j))$.

In kernel space, positive samples are clustered by kernel Ward's linkage clustering (Yeung et al., 2007). In the zth view, the covariance matrix of cluster $P_c^{(v)}$ is expressed as

$$\boldsymbol{\Sigma}_{P_c^{(v)}}^{(z)} = \frac{1}{|P_c^{(v)}|} \hat{\boldsymbol{S}}^\top \hat{\boldsymbol{S}}, \tag{42}$$

where $\hat{\boldsymbol{S}} = \Phi(\boldsymbol{X}_{P_c^{(v)}}^{(z)}) - \Phi(\boldsymbol{M}_{P_c^{(v)}}^{(z)})$. $\boldsymbol{X}_{P_c^{(v)}}^{(z)}$ is a sample matrix corresponding to cluster $P_c^{(v)}$. $\boldsymbol{M}_{P_c^{(v)}}^{(z)}$ is the mean matrix of $\boldsymbol{X}_{P_c^{(v)}}^{(z)}$, and each row of it is $\boldsymbol{\mu}_{P_c^{(v)}}^{(z)}$. The definition of $|P_c^{(v)}|$, $P_c^{(v)}$, $c_+^{(v)}$, $\boldsymbol{\mu}_{P_c^{(v)}}^{(z)}$ are shown in Table 1. Thus, we obtain

$$\Phi(\boldsymbol{C}_v)^\top \boldsymbol{\Sigma}_{P_c^{(v)}}^{(z)} \Phi(\boldsymbol{C}_v) = \sum_{c=1}^{c_+^{(v)}} \frac{1}{|P_c^{(v)}|} \hat{\boldsymbol{Q}}^\top \hat{\boldsymbol{Q}}, \tag{43}$$

where $\hat{\boldsymbol{Q}} = K(\boldsymbol{X}_{P_c^{(v)}}^{(z)}, \boldsymbol{C}_v) - K(\boldsymbol{M}_{P_c^{(v)}}^{(z)}, \boldsymbol{C}_v)$.

Similar to Eqs. (13) and (14), the total covariance matrices of positive and negative class are formulated respectively as

$$\hat{\boldsymbol{\Sigma}}_+^{(z)} = \sum_{v=1}^{V} \theta_+^{(v)} \sum_{c=1}^{c_+^{(v)}} \boldsymbol{\Sigma}_{P_c^{(v)}}^{(z)} \quad \text{and} \quad \hat{\boldsymbol{\Sigma}}_-^{(z)} = \sum_{v=1}^{V} \theta_-^{(v)} \sum_{c=1}^{c_-^{(v)}} \boldsymbol{\Sigma}_{N_c^{(v)}}^{(z)}, \tag{44}$$

where $\theta_+^{(v)}, \theta_-^{(v)}$ represent the view weight.

When the view weight is fixed, we can receive the two kernel-generated hyperplanes by solving the following two problems:

$$\min_{\boldsymbol{w}_+^{(v)}, \theta_+, \xi_+^{(v)}} \quad \frac{1}{2} \sum_{v=1}^{V} \theta_+^{(v)} \left\| \boldsymbol{X}_+^{(v)} \boldsymbol{w}_+^{(v)} \right\|^2 + \frac{1}{2} \sum_{v=1}^{V} \theta_+^{(v)} \left\| \boldsymbol{w}_+^{(v)} \right\|^2 + \frac{\lambda}{2} \|\theta_+\|^2 + C_+ \sum_{v=1}^{V} \boldsymbol{e}_-^\top \boldsymbol{\xi}_+^{(v)} + \frac{D_+}{2} \sum_{v=1}^{V} \boldsymbol{w}_+^{(v)\top} \boldsymbol{\Sigma}_+^{(v)} \boldsymbol{w}_+^{(v)}$$
$$s.t. \quad -\boldsymbol{X}_-^{(v)} \boldsymbol{w}_+^{(v)\top} \ge \boldsymbol{e}_- - \boldsymbol{\xi}_+^{(v)}, \boldsymbol{\xi}_+^{(v)} \ge \mathbf{0},$$
$$\sum_{v=1}^{V} \theta_+^{(v)} = 1, \theta_+^{(v)} \ge 0, v = 1, \dots, V \tag{45}$$

and

$$\begin{aligned}\min_{\boldsymbol{w}_-^{(v)},\boldsymbol{\theta}_-,\boldsymbol{\xi}_-^{(v)}} \quad & \frac{1}{2}\sum_{v=1}^{V}\theta_-^{(v)}\left\|\boldsymbol{X}_-^{(v)}\boldsymbol{w}_-^{(v)}\right\|^2+\frac{1}{2}\sum_{v=1}^{V}\theta_-^{(v)}\left\|\boldsymbol{w}_-^{(v)}\right\|^2+\frac{\eta}{2}\|\boldsymbol{\theta}_+\|^2\\ & +C_-\sum_{v=1}^{V}\boldsymbol{e}_+^{\top}\boldsymbol{\xi}_-^{(v)}+\frac{D_-}{2}\sum_{v=1}^{V}\boldsymbol{w}_-^{(v)\top}\boldsymbol{\Sigma}_-^{(v)}\boldsymbol{w}_-^{(v)}\\ s.t. \quad & \boldsymbol{X}_+^{(v)}\boldsymbol{w}_-^{(v)\top}\geq\boldsymbol{e}_+-\boldsymbol{\xi}_-^{(v)},\boldsymbol{\xi}_-^{(v)}\geq\boldsymbol{0},\\ & \sum_{v=1}^{V}\theta_-^{(v)}=1,\theta_-^{(v)}\geq 0,v=1,\ldots,V,\end{aligned}\tag{46}$$

where $\boldsymbol{\Sigma}_+^{(v)}=\begin{bmatrix}\boldsymbol{\Phi}(\boldsymbol{C}_v)^{\top}\hat{\boldsymbol{\Sigma}}_+^{(v)}\boldsymbol{\Phi}(\boldsymbol{C}_v) & \boldsymbol{0}\\ \boldsymbol{0} & 0\end{bmatrix}$ and $\boldsymbol{\Sigma}_-^{(v)}=\begin{bmatrix}\boldsymbol{\Phi}(\boldsymbol{C}_v)^{\top}\hat{\boldsymbol{\Sigma}}_+^{(v)}\boldsymbol{\Phi}(\boldsymbol{C}_v) & \boldsymbol{0}\\ \boldsymbol{0} & 0\end{bmatrix}$, $C_+, C_-, D_+, D_-, \lambda, \eta$ are predefined tradeoff factors and $\boldsymbol{\xi}_+^{(v)}, \boldsymbol{\xi}_-^{(v)}$ are slack variables for two hyperplanes in the vth view. Afterward, we can get the classifier parameters

$$\boldsymbol{w}_+^{(v)}=-\left(\theta_+^{(v)}\boldsymbol{X}_+^{(v)\top}\boldsymbol{X}_+^{(v)}+\theta_+^{(v)}\boldsymbol{I}_++D_+\boldsymbol{\Sigma}_+^{(v)}\right)^{-1}\boldsymbol{X}_-^{(v)\top}\boldsymbol{\alpha}_+^{(v)},\tag{47}$$

$$\boldsymbol{w}_-^{(v)}=\left(\theta_-^{(v)}\boldsymbol{X}_-^{(v)\top}\boldsymbol{X}_-^{(v)}+\theta_-^{(v)}\boldsymbol{I}_-+D_-\boldsymbol{\Sigma}_-^{(v)}\right)^{-1}\boldsymbol{X}_+^{(v)\top}\boldsymbol{\alpha}_-^{(v)}.\tag{48}$$

Thus, the dual of problems (45) and (46) are

$$\begin{aligned}\min_{\{\boldsymbol{\alpha}_+^{(v)}\}_{v=1}^{V}} \quad & \sum_{v=1}^{V}\left(\frac{1}{2}\boldsymbol{\alpha}_+^{(v)\top}\boldsymbol{L}_+^{(v)}\boldsymbol{\alpha}_+^{(v)}-\boldsymbol{e}_-^{\top}\boldsymbol{\alpha}_+^{(v)}\right)\\ s.t. \quad & \boldsymbol{0}\leq\boldsymbol{\alpha}_+^{(v)}\leq C_+\boldsymbol{e}_-,v=1,\ldots,V,\end{aligned}\tag{49}$$

and

$$\begin{aligned}\min_{\{\boldsymbol{\alpha}_-^{(v)}\}_{v=1}^{V}} \quad & \sum_{v=1}^{V}\left(\frac{1}{2}\boldsymbol{\alpha}_-^{(v)\top}\boldsymbol{L}_-^{(v)}\boldsymbol{\alpha}_-^{(v)}-\boldsymbol{e}_+^{\top}\boldsymbol{\alpha}_-^{(v)}\right)\\ s.t. \quad & \boldsymbol{0}\leq\boldsymbol{\alpha}_-^{(v)}\leq C_-\boldsymbol{e}_+,v=1,\ldots,V,\end{aligned}\tag{50}$$

where

$$\boldsymbol{L}_+^{(v)}=\boldsymbol{X}_-^{(v)}(\theta_+^{(v)}\boldsymbol{X}_+^{(v)\top}\boldsymbol{X}_+^{(v)}+\theta_+^{(v)}\boldsymbol{I}_++D_+\boldsymbol{\Sigma}_+^{(v)})^{-1}\boldsymbol{X}_-^{(v)\top},\tag{51}$$

$$\boldsymbol{L}_-^{(v)}=\boldsymbol{X}_+^{(v)}(\theta_-^{(v)}\boldsymbol{X}_-^{(v)\top}\boldsymbol{X}_-^{(v)}+\theta_-^{(v)}\boldsymbol{I}_-+D_-\boldsymbol{\Sigma}_-^{(v)})^{-1}\boldsymbol{X}_+^{(v)\top},\tag{52}$$

and $\boldsymbol{\alpha}_+^{(v)}$, $\boldsymbol{\alpha}_-^{(v)}$ are Lagrange multiplier vectors, $\boldsymbol{I}_+, \boldsymbol{I}_-$ are identity matrices of proper dimensions.

Similar to linear case, given a multi-view test sample $\boldsymbol{x}$, its label is decided by the following function

$$f(\boldsymbol{x})=sgn(\sum_{v=1}^{V}\theta_+^{(v)}|[K(\boldsymbol{x}^{(v)\top},\boldsymbol{C}_v^{\top})\ 1]\boldsymbol{w}_+^{(v)}|-\sum_{v=1}^{V}\theta_-^{(v)}|[K(\boldsymbol{x}^{(v)\top},\boldsymbol{C}_v^{\top})\ 1]\boldsymbol{w}_-^{(v)}|).\tag{53}$$

The algorithm of nonlinear MvSTSVM is concluded in Algorithm 2.

Algorithm 2 Nonlinear MvSTSVM

1: **Input**: training set $\mathcal{X}=\{(\boldsymbol{x}_i^{(1)},\ldots,\boldsymbol{x}_i^{(V)},y_i)\}_{i=1}^{n}$, parameters $(C_+,C_-,D_+,D_-,\lambda,\eta)$, kernel function K, maximum iterations t_{max};
2: **Initialize**: $\theta_+^0=\frac{1}{V},\theta_-^0=\frac{1}{V},t=t'=0$;
3: **While** $\|\theta_+^{t+1}-\theta_+^t\|>\delta$ and $t\leq t_{max}$ **do**
4: acquire $\boldsymbol{w}_+^t$ by solving dual problem (49);
5: acquire θ_+^{t+1} according to the optimization model (31);
6: $t=t+1$;
7: **End While**
8: **While** $\|\theta_-^{t'+1}-\theta_-^{t'}\|>\delta$ and $t'\leq t_{max}$ **do**
9: acquire $\boldsymbol{w}_-^{t'}$ by solving dual problem (50);
10: acquire $\theta_-^{t'+1}$ according to the optimization model (32);
11: $t'=t'+1$;
12: **End While**
13: **Output**: given a test sample $\boldsymbol{x}$, its label y depends on predictor (53).

3.3. Discussion

In the following subsection, we discuss the connection among MvSTSVM, MvTSVM (Xie & Sun, 2015), MvTSVM2C (Xie & Sun, 2019) and MvSLMC (Zhao & Xu, 2023), respectively. Moreover, the computational complexity of MvSTSVM is also analyzed based on its two optimization problems.

3.3.1. MvSTSVM vs. MvTSVM

(1) MvSTSVM is similar to MvTSVM in structure and inherits all advantages of MvTSVM. Even better, MvSTSVM extends MvTSVM from two-view to multi-view, which can make our model more universal. Meanwhile, MvSTSVM makes full use of each view as weighting information instead of splitting the perspective weights equally like MvTSVM. Thus, MvSTSVM tends to have higher accuracy in classification tasks.

(2) Clustering technique is utilized in MvSTSVM to capture the structural information within classes and views. Based on weight idea and data distribution information, MvSTSVM is guided by both consensus and complementarity principles. However, MvTSVM only follows the principle of consensus because it does not consider the relationship between different perspectives.

(3) MvTSVM and MvSTSVM achieve their models by solving QPPs for positive and negative hyperplanes. The difference is that MvTSVM solves two dual problems directly while MvSTSVM uses an iterative algorithm to deal with them.

3.3.2. MvSTSVM vs. MvTSVM2C

(1) MvSTSVM and MvTSVM2C both obtain the classifiers by finding nonparallel hyperplanes. MvTSVM2C constructs two global hyperplanes resulting in complex quadratic matrices, while MvSTSVM solves two small QPPs in each view thus the solution of the problem is accelerated.

(2) Same as MvTSVM2C, our model also follows consensus and complementarity principles. MVTSVM2C achieves its consensus principle by minimizing the gap of hyperplanes among different perspectives. MvSTSVM keeps consistency information in sample clustering.

(3) Both MvSTSVM and MvTSVM2C can perform multi-view tasks and simultaneously adopt the weighting idea. In MvTSVM2C, each view shares the same weight in both hyperplanes. In MvSTSVM, the view weights in two optimization problems are independent rather than shared, which makes the solution of our model more flexible.

3.3.3. MvSTSVM vs. MvSLMC

(1) As an improved version of SVM-2K, MvSLMC constructs a large QPP for solution, while our model is implemented by solving two small QPPs. Consequently, MvSTSVM has better training efficiency than MvSLMC especially when dealing with large datasets.

(2) Kernel method leads to the separation of linearly inseparable samples. However, according to the work of Zhao and Xu (2023), MvSLMC cannot use a kernel trick, resulting in poor performance in the classification of linearly indivisible datasets. As for MvSTSVM, we give the nonlinear pattern by mapping a low-dimensional sample to a high-dimensional space, which makes our model more competitive than MvSLMC.

(3) Like MvSLMC, MvSTSVM adds the view weight into the object function and preserves more connections within distinct views. Besides, they satisfy both principles of consensus and complementarity. The difference is that MvSLMC focuses on the structural information of two classes simultaneously, whereas MvSTSVM considers this information of which the hyperplane is nearer to the class in each model. It enables MvSTSVM to collect more effective prior knowledge and perform better in classification.

3.3.4. Algorithmic complexity analysis

In linear MvSTSVM, let $l = n_1 = n_2$. p is the total dimension of data and p_v denotes the dimension of the vth view. The perspective weights converge after t iterations. The complexity of the MvSTSVM is mainly affected by clustering, solving QP problem and matrix inversion.

In the clustering session, samples of two classes are merged independently. It takes some time to calculate the distance between clusters. When the positive samples are clustered, the complexity is $\mathcal{O}(l^2 p)$. Likewise, the algorithmic complexity of the negative one is $\mathcal{O}(l^2 p)$.

In terms of QP problem solution and matrix inversion, they are implemented for each QPP in each perspective. The dual formulations (27) and (28) for obtaining $\{\boldsymbol{w}_{\pm}^{(v)}\}_{v=1}^{V}$ involve a quadratic matrix with size $l \times l$. Hence, the algorithmic complexity is $\mathcal{O}(Vl^3)$. In addition, the size of two inverse matrices in each view is both $p_v \times p_v$ (see Eq. (29)). Therefore, the computational complexity of matrix inversion is $\mathcal{O}(p^3)$. The computational complexity of obtaining $\{\theta_{\pm}^{(v)}\}_{v=1}^{V}$ can be ignored.

In summary, the algorithmic complexity of MvSTSVM is $\mathcal{O}(2l^2 p + 2t(Vl^3 + p^3))$.

4. Safe screening rules for MvSTSVM

In this part, some safe screening rules for MvSTSVM (SSR-MvSTSVM) will be deduced to speed up the training process.

According to problem (18), we classify the negative training samples into three sets:

$$\begin{cases} \mathcal{R} = \{i \mid -\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} > 1\}, i = 1, \dots, n_2, \\ \mathcal{L} = \{i \mid -\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} < 1\}, i = 1, \dots, n_2, \\ \mathcal{E} = \{i \mid -\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} = 1\}, i = 1, \dots, n_2, \end{cases} \tag{54}$$

where $\mathcal{R}, \mathcal{L}, \mathcal{E}$ represent those samples outside, inside and on the interval boundaries of the hyperplane, respectively. For an arbitrary sample $\boldsymbol{x}_{-,i}^{(v)}, i = 1, \dots, n_2$, we can get the following derivation based on KKT condition (21):

$$\begin{cases} -\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} > 1 \Rightarrow \xi_{+,i}^{(v)} = 0 \Rightarrow \alpha_{+,i}^{(v)} = 0, \\ -\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} < 1 \Rightarrow \xi_{+,i}^{(v)} > 0 \Rightarrow \beta_{+,i}^{(v)} = 0 \Rightarrow \alpha_{+,i}^{(v)} = C_+, \\ -\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} = 1 \Rightarrow \xi_{+,i}^{(v)} = 0 \Rightarrow \alpha_{+,i}^{(v)} \in [0, C_+], \end{cases} \tag{55}$$

that is to say,

$$\begin{cases} i \in \mathcal{R} \Rightarrow \alpha_{+,i}^{(v)} = 0, \\ i \in \mathcal{L} \Rightarrow \alpha_{+,i}^{(v)} = C_+, \\ i \in \mathcal{E} \Rightarrow \alpha_{+,i}^{(v)} \in [0, C_+]. \end{cases} \tag{56}$$

Since those $\boldsymbol{x} \in \{\boldsymbol{x}_{-,i}^{(v)} \mid i \in \mathcal{R} \cup \mathcal{L}\}$ do not affect the final classifier, they are named non-support vectors (non-SVs). And $\boldsymbol{x} \in \{\boldsymbol{x}_{-,i}^{(v)} \mid i \in \mathcal{E}\}$ are support vectors (SVs). In terms of current research, most of the samples are non-SVs and only a small part of them are SVs during training (Pan, Yang, Xu, & Wang, 2017; Wang et al., 2023). If we can identify those non-SVs in advance and remove them from the QPPs, the training efficiency will be obviously improved. To obtain sets $\mathcal{R}$ and $\mathcal{L}$, we can first estimate the lower and upper bound of $-\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)}$. Once the subsets $\mathcal{R}' \subset \mathcal{R}$ and $\mathcal{L}' \subset \mathcal{L}$ are found, the non-SVs are determined. Firstly, we give an introduction to the variational inequality as the following lemma.

Lemma 1. *Suppose $A \in \mathbb{R}^n$ is a convex set and $F(\boldsymbol{x})$ is a Gâteaux differentiable function defined on an open set containing A, then $\forall \boldsymbol{x} \in A$,*

$$\langle \nabla F(\boldsymbol{x}^*), \boldsymbol{x} - \boldsymbol{x}^* \rangle \geq 0, \tag{57}$$

where $\boldsymbol{x}^*$ is a local minimizer of $F(\boldsymbol{x})$ on A.

Assume that D_+, λ are fixed, $\{\boldsymbol{w}_{+}^{k(v)}\}_{v=1}^{V}$ and $\{\theta_{+}^{k(v)}\}_{v=1}^{V}$ are the best solutions to the QP problem (16) when parameter $C_+ = C_+^k$. $\boldsymbol{\alpha}_{+}^{k(v)}$ is the Lagrange multiplier vector corresponding to $\boldsymbol{w}_{+}^{k(v)}$. In dual problem (27), C_+ is a parameter that needs to search. We deduce a screening rule to accelerate this process in the following theorem.

Theorem 1. *Define $\boldsymbol{R}_{+}^{(v)} = \boldsymbol{M}^{(v)} \boldsymbol{M}^{(v)\top}$, for problem (27), given a set of parameters in the sequence $0 < C_+^1 < C_+^2 < \cdots < C_+^K$ and $C_+^{k+1} (0 < k < K)$ is the current parameter. In the vth view, $\boldsymbol{\alpha}_{+}^{k(v)}$ is known. Then for $\forall i = 1, \dots, n_2$, if*

$$\frac{1}{2} \boldsymbol{M}_i^{(v)} \boldsymbol{M}^{(v)\top} \frac{C_+^k + C_+^{k+1}}{C_+^k} \alpha_{+,i}^{k(v)} - \frac{1}{2} \left\| \boldsymbol{M}_i^{(v)} \right\| \left\| \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \alpha_{+,i}^{k(v)} \right\| > 1, \tag{58}$$

then $\alpha_{+,i}^{k+1(v)} = 0$, $(i \in \mathcal{R})$ holds, and if

$$\frac{1}{2} \boldsymbol{M}_i^{(v)} \boldsymbol{M}^{(v)\top} \frac{C_+^k + C_+^{k+1}}{C_+^k} \alpha_{+,i}^{k(v)} + \frac{1}{2} \left\| \boldsymbol{M}_i^{(v)} \right\| \left\| \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \alpha_{+,i}^{k(v)} \right\| < 1, \tag{59}$$

then $\alpha_{+,i}^{k+1(v)} = C_+^{k+1}$, $(i \in \mathcal{L})$ holds.

Proof. In dual formulation (27), its object function $F(\boldsymbol{\alpha}_{+}^{k(v)})$ is defined on a convex set $A_k = \{\boldsymbol{0} \leq \boldsymbol{\alpha}_{+}^{k(v)} \leq C_+^k \boldsymbol{e}_-\}$. Inequality (57) implies that

$$\langle \nabla F(\boldsymbol{\alpha}_{+}^{k(v)}), \hat{\boldsymbol{\alpha}}_{+}^{k(v)} - \boldsymbol{\alpha}_{+}^{k(v)} \rangle \geq 0, \ \forall \hat{\boldsymbol{\alpha}}_{+}^{k(v)} \in A_k, \tag{60}$$

$$\langle \nabla F(\boldsymbol{\alpha}_{+}^{k+1(v)}), \hat{\boldsymbol{\alpha}}_{+}^{k+1(v)} - \boldsymbol{\alpha}_{+}^{k+1(v)} \rangle \geq 0, \ \forall \hat{\boldsymbol{\alpha}}_{+}^{k+1(v)} \in A_{k+1}, \tag{61}$$

where $\nabla F(\boldsymbol{\alpha}_{+}^{k(v)}) = \boldsymbol{R}_{+}^{(v)} \boldsymbol{\alpha}_{+}^{k(v)} - \boldsymbol{e}_-$.

We can easily find that $(C_+^k / C_+^{k+1}) \boldsymbol{\alpha}_{+}^{k+1(v)} \in A_k$ and $(C_+^{k+1} / C_+^k) \boldsymbol{\alpha}_{+}^{k(v)} \in A_{k+1}$. Hence, we can define $\hat{\boldsymbol{\alpha}}_{+}^{k(v)} = (C_+^k / C_+^{k+1}) \hat{\boldsymbol{\alpha}}_{+}^{k+1(v)}$ and $\hat{\boldsymbol{\alpha}}_{+}^{k+1(v)} = (C_+^{k+1} / C_+^k) \hat{\boldsymbol{\alpha}}_{+}^{k(v)}$ so that they are located in the convex set A_k and A_{k+1} respectively.

Therefore, the variational inequalities (60) and (61) become

$$\langle \boldsymbol{R}_{+}^{(v)} \boldsymbol{\alpha}_{+}^{k(v)} - \boldsymbol{e}_-, \frac{C_+^k}{C_+^{k+1}} \boldsymbol{\alpha}_{+}^{k+1(v)} - \boldsymbol{\alpha}_{+}^{k(v)} \rangle \geq 0, \tag{62}$$

$$\langle \boldsymbol{R}_{+}^{(v)} \boldsymbol{\alpha}_{+}^{k(v)} - \boldsymbol{e}_-, \frac{C_+^{k+1}}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} - \boldsymbol{\alpha}_{+}^{k+1(v)} \rangle \geq 0. \tag{63}$$

Adding inequalities (62) and (63) together and we obtain

$$\left\| \boldsymbol{M}^{(v)\top} \boldsymbol{\alpha}_{+}^{k+1(v)} - \frac{1}{2} \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} + C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} \right\| \leq \frac{1}{2} \left\| \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} \right\|. \tag{64}$$

If a sample $\boldsymbol{x}_{-,i}^{(v)} (i = 1, \dots, n_2)$ satisfies $-\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} > 1$, we can derive that $\boldsymbol{R}_{+}^{(v)} \alpha_{+,i}^{(v)} > 1$. Then $\alpha_{+,i}^{(v)} = 0$ holds. Similarly, if the sample satisfies $-\boldsymbol{w}_{+}^{(v)} \boldsymbol{x}_{-,i}^{(v)} < 1$, then $\boldsymbol{R}_{+}^{(v)} \alpha_{+,i}^{(v)} < 1$ holds. Then $\alpha_{+,i}^{(v)} = C_+$ holds. Now our purpose translates into estimating the lower and upper bound of $\boldsymbol{R}_{+}^{(v)} \alpha_{+,i}^{(v)}$.

According to inequality (64), we have

$$\begin{aligned} \boldsymbol{R}_{+,i}^{(v)} \boldsymbol{\alpha}_{+}^{k+1(v)} &= \boldsymbol{M}_i^{(v)} \boldsymbol{M}^{(v)\top} \boldsymbol{\alpha}_{+}^{k+1(v)} \\ &= \boldsymbol{M}_i^{(v)} (\boldsymbol{M}^{(v)\top} \boldsymbol{\alpha}_{+}^{k+1(v)} - \frac{1}{2} \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} + C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)}) \\ &\quad + \frac{1}{2} \boldsymbol{M}_i^{(v)} \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} \\ &\geq - \| \boldsymbol{M}_i^{(v)} \| \| \boldsymbol{M}^{(v)\top} \boldsymbol{\alpha}_{+}^{k+1(v)} - \frac{1}{2} \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} + C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} \| \\ &\quad + \frac{1}{2} \boldsymbol{M}_i^{(v)} \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} \\ &\geq - \frac{1}{2} \| \boldsymbol{M}_i^{(v)} \| \| \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)} \| \\ &\quad + \frac{1}{2} \boldsymbol{M}_i^{(v)} \boldsymbol{M}^{(v)\top} \frac{C_+^{k+1} - C_+^k}{C_+^k} \boldsymbol{\alpha}_{+}^{k(v)}, \end{aligned} \tag{65}$$

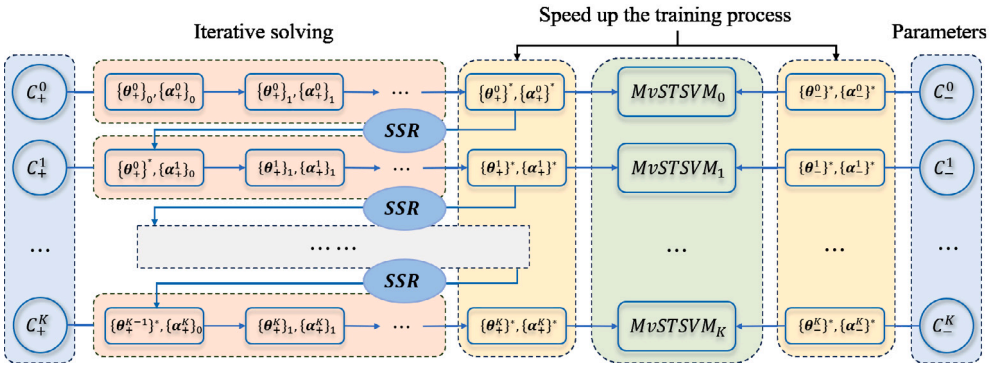


Fig. 3. The schematic diagram of SSR-MvSTSVM.

and

$$\begin{aligned}\boldsymbol{R}_{+,i}^{(v)}\boldsymbol{\alpha}_{+}^{k+1(v)} &\le \|\boldsymbol{M}_i^{(v)}\|\|\boldsymbol{M}^{(v)\top}\boldsymbol{\alpha}_{+}^{k+1(v)} - \frac{1}{2}\boldsymbol{M}^{(v)\top}\frac{C_{+}^{k+1}+C_{+}^{k}}{C_{+}^{k}}\boldsymbol{\alpha}_{+}^{k(v)}\| \\ &\quad + \frac{1}{2}\boldsymbol{M}_i^{(v)}\boldsymbol{M}^{(v)\top}\frac{C_{+}^{k+1}-C_{+}^{k}}{C_{+}^{k}}\boldsymbol{\alpha}_{+}^{k(v)} \\ &\le \frac{1}{2}\|\boldsymbol{M}_i^{(v)}\|\|\boldsymbol{M}^{(v)\top}\frac{C_{+}^{k+1}-C_{+}^{k}}{C_{+}^{k}}\boldsymbol{\alpha}_{+}^{k(v)}\| \\ &\quad + \frac{1}{2}\boldsymbol{M}_i^{(v)}\boldsymbol{M}^{(v)\top}\frac{C_{+}^{k+1}-C_{+}^{k}}{C_{+}^{k}}\boldsymbol{\alpha}_{+}^{k(v)}, \end{aligned} \tag{66}$$

where $\boldsymbol{M}_i^{(v)}$ is the i row of matrix $\boldsymbol{M}^{(v)}$.

In this way, we can obtain inequalities (58) and (59) as the safe screening rules for negative samples in MvSTSVM. Now the following safe screening rules are designed for positive samples in MvSTSVM.

Theorem 2. *Define $\boldsymbol{R}_{-}^{(v)} = \boldsymbol{N}^{(v)}\boldsymbol{N}^{(v)\top}$. For problem(28), given a set of parameters in sequence $0 < C_{-}^{1} < C_{-}^{2} < \cdots < C_{-}^{K}$ and $C_{-}^{k+1}(0 < k < K)$ is the current parameter. In the vth view, $\boldsymbol{\alpha}_{-}^{k(v)}$ is known. Then for $\forall i = 1, \ldots, n_1$, if*

$$\frac{1}{2}\boldsymbol{N}_i^{(v)}\boldsymbol{N}^{(v)\top}\frac{C_{-}^{k}+C_{-}^{k+1}}{C_{-}^{k}}\alpha_{-,i}^{k(v)} - \frac{1}{2}\left\|\boldsymbol{N}_i^{(v)}\right\|\left\|\boldsymbol{N}^{(v)\top}\frac{C_{-}^{k+1}-C_{-}^{k}}{C_{-}^{k}}\alpha_{-,i}^{k(v)}\right\| > 1, \tag{67}$$

then $\alpha_{-,i}^{k+1(v)} = 0$, $(i \in \mathcal{R})$ holds and if

$$\frac{1}{2}\boldsymbol{N}_i^{(v)}\boldsymbol{N}^{(v)\top}\frac{C_{-}^{k}+C_{-}^{k+1}}{C_{-}^{k}}\alpha_{-,i}^{k(v)} + \frac{1}{2}\left\|\boldsymbol{N}_i^{(v)}\right\|\left\|\boldsymbol{N}^{(v)\top}\frac{C_{-}^{k+1}-C_{-}^{k}}{C_{-}^{k}}\alpha_{-,i}^{k(v)}\right\| < 1, \tag{68}$$

then $\alpha_{-,i}^{k+1(v)} = C_{-}^{k+1}$, $(i \in \mathcal{L})$ holds.

The process of SSR-MvSTSVM is depicted in Fig. 3. $\{\cdot\}_t$ denotes the solution in the tth iterations. For example, when parameter C_+ is fixed at C_+^0, $\{\theta_+^0\}_1$ stands for the positive view weight in the first iteration, and the solution converges until the view weight updates to $\{\theta_+^0\}^*$.

5. Numerical experiments

In this section, a series of experiments are done to confirm the excellent performance of our model. Our experiments are performed on six datasets including 4 real-world datasets and a synthetic dataset. Basic information about them is shown in Table 2. All data are standardized before they are conducted. In addition, our experiments are implemented in MATLAB 2021b on Windows 7 running on a PC with system configuration Intel(R) Core(TM) i5-7500 CPU(3.40 GHz) and 4.00 GB of RAM. Meanwhile, SSR is applied to improve computing efficiency.

5.1. Experimental settings

Datasets.

(1) *UCI datasets*[2]

UCI Machine Learning Repository is a classical dataset in machine learning. We select two single-view datasets: Haberman and Mushroom. Their dimensions are 3 and 4, respectively. The second view is derived from the PCA technique.

(2) *AwA*[3]

Animal with Attributes(AwA) is a dataset containing 30475 images of animals from 50 species. We pick ten kinds of animals (beaver, blue+whale, skunk, ox, spider+monkey, rat, walrus, weasel, mouse, mole) from AwA and pair them into 45 binary datasets based on *one vs. one* strategy. In addition, our AwA datasets include two views: Global Color Histogram (CQ) and Scale Invariant Feature Transform (SIFT). After extracting 95% of the principal components from the original views, the dimensions of each view is 653 and 252, respectively.

(3) *CIFAR-10*[4]

The CIFAR-10 dataset is composed of 60000 color images with the size of 32 × 32 from 10 classes (6000 images per class)—airplane, automobile, cat, bird, deer, dog, horse, frog, ship, truck. For each class, the first 1000 samples are chosen. We combine these 10 classes by *one vs. one* strategy to get 45 binary datasets. Afterward, 4 features are extracted for 4 views. The feature sets and their dimensions are (CQ, 252), (Gabor, 80), (HOG, 324) and (LBP, 59).

(4) *Synthetic dataset*[5]

This artificial dataset is selected from three-view synthetic dataset "synthetic1" in article (Wang et al., 2023). The dimensions of all three views are 3. More information about this dataset can be found in the "Numerical experiments" of article (Wang et al., 2023). The data distribution is shown in Fig. 4. They are view A, view B and view C from left to right.

Kernels. The Gaussian radial basis function (RBF) is widely used in binary classification. However, the comparison model MvSLMC in our experiment cannot use a kernel trick. Therefore, RBF is no longer applicable. In order to validate the reasonableness of the experimental results, we uniformly use a linear kernel $K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{x}_i^{\top} \cdot \boldsymbol{x}_j$ on all models.

Parameters. We make comparisons between MvSTSVM with five related models, including SVM-2K, MvSVM2C, MvTSVM, MvTSVM2C and MvSLMC. To obtain the optimal parameter, the grid search is adopted for all models. Meanwhile, five-fold-cross-validation is implemented. All results are the average results of five experiments with

[2] https://archive.ics.uci.edu/ml/index.php
[3] https://cvml.ist.ac.at/AwA
[4] http://www.cs.toronto.edu/~kriz/cifar.html
[5] https://github.com/GlassBALDY/SSR-MvSVMs/tree/master/Data

Table 2
Information of used datasets.

Datasets	Harberman	Mushroom	AWA	CIFAR-10	Sythetic dataset
View 1	Original view(3)	Original view(4)	CQ(653)	CQ(252)	A(22)
View 2	Extracted view(3)	Extracted view(1)	SIFT(252)	Gabor(80)	B(11)
View 3	–	–	–	HOG(324)	C(84)
View 4	–	–	–	LBP(59)	–
Samples	306	1559	2469	10 000	800
Classes	2	2	2	4	3

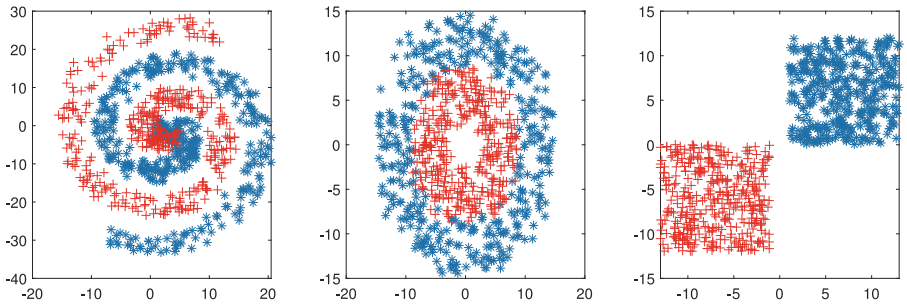


Fig. 4. Illustration of synthetic dataset.

the same parameter but different testing sets. For SVM-2K, MvSLMC and MvSVM2C, all penalty parameters are chosen from sequence $\{10^i | i = -3 : 1 : 3\}$. In MvTSVM, let $C_+ = C_-, D_+ = D_-$, which are searched from $\{10^i | i = -3 : 1 : 3\}$. In MvTSVM2C, we let $C = C_1 = C_2$ and C, D, ρ are turned over $\{10^i | i = -3 : 1 : 3\}$. For MvSTSVM, we set $C_+ = C_-, D_+ = D_-, \lambda = \eta$ and select them from $\{10^i | i = -3 : 1 : 3\}$. Besides, the insensitive parameter ϵ in MvTSVM, SVM-2K and MvSVM2C is set to 0.1. For simplicity, the kernel parameters in all methods are fixed at $\sigma = 1$.

5.2. Experiments on two-view datasets

We make comparisons between MvSTSVM with other benchmark algorithms on two-view datasets. All experiment results are documented in Tables 3 and 4.

In Table 3, the last two rows includes the average accuracy, time and rank among 12 datasets. The bold values represent the best accuracy of each row. It is evident that our proposed MvSTSVM behaves more excellently compared to other benchmark algorithms in both average and rank. Our method obtains the highest accuracy in general, i.e., 26 out of 45 times for two-view datasets. Although MvSTSVM cannot perform the best on certain datasets, the accuracy is similar to the best results.

In addition to accuracy, AUC, F1-score and precision are critical for providing a more comprehensive evaluation of classifier performance. We record the values for these metrics in Table 4, with the best results in each row marked in bold for clarity. Our method achieves consistently excellent results above these metrics, which demonstrates its robust and reliable classification performance, not only in terms of overall accuracy, but also in terms of strong predictive power in effectively distinguishing between classes. This comprehensive assessment further shows the superiority of our method.

The scatter diagram in Fig. 5 depicts the training time of our proposed MvSTSVM and five compared algorithms on AwA datasets. Each algorithm is distinguished by different colors and shapes. For all datasets, MvSTSVM is comparable and efficient with less time consumption. According to the average time, the model with two hyperplanes takes less time to train than those with a single hyperplane. SVM2K and MvSVM2C embed the similarity constraint between two distinct views into the restrictive condition, which is more complex and time-consuming than our method. Therefore, they take the most time on all two-view datasets.

5.3. Experiments on multi-view datasets

Table 5 lists the highest accuracy and the corresponding time of multi-view classifiers on CIFAR-10 and the synthetic dataset. The last two rows of Table 5 record the average accuracy and rank for each model. Besides, AUC, F1-score and precision are adopted to evaluate all models comprehensively. The value of these metrics are shown in Table 6. The best performance of the same dataset is marked in boldface.

To provide a more clear comparison of three multi-view models on CIFAR-10, the information of the winning algorithm on each dataset is shown in Fig. 6. Different algorithms are distinguished by colors. The height of each bar represents the accuracy of the winning algorithm. From the results, it is evident that our proposed MvSTSVM achieves the highest accuracy across 45 datasets in most cases, surpassing both MvSLMC and MvTSVM2C. Moreover, our model is more comparable and efficient on all 45 classification tasks. With a higher computational complexity during QPPs, MvSLMC consumes the most time in the experiment.

On the synthetic dataset, the best accuracy of our proposed algorithm is 100% followed by MvSLMC (99.25%) and MvTSVM2C (98.47%). Therefore, MvSTSVM obtains the highest accuracy of 100%. This remarkable result indicates that MvSTSVM successfully distinguishes between positive and negative samples on synthetic dataset, further indicating its robustness and superior generalization capability.

Among multi-view datasets, MvSTSVM performs better compared to other multi-view classifiers both on accuracy as well as average processing time. This superiority stems from its unique approach to leveraging structural information within inter-class and intra-class through a clustering method, thereby maintaining the consistency information across distinct views. Besides, MvSTSVM integrates the weight idea into positive and negative classes independently, which enables the solution more flexible.

5.4. Demonstration of the safe screening process

To clearly demonstrate the effectiveness of our proposed acceleration algorithm SSR-MvSTSVM, we present the safe screening process in Fig. 7, conducted on six different datasets: cifar1, cifar2, cifar3, cifar4, Mushroom and the sythetic dataset.

In the first subplot, we plot the training time and screening ratio for SSR-MvSTSVM and Raw-MvSTSVM according to parameter sequence.

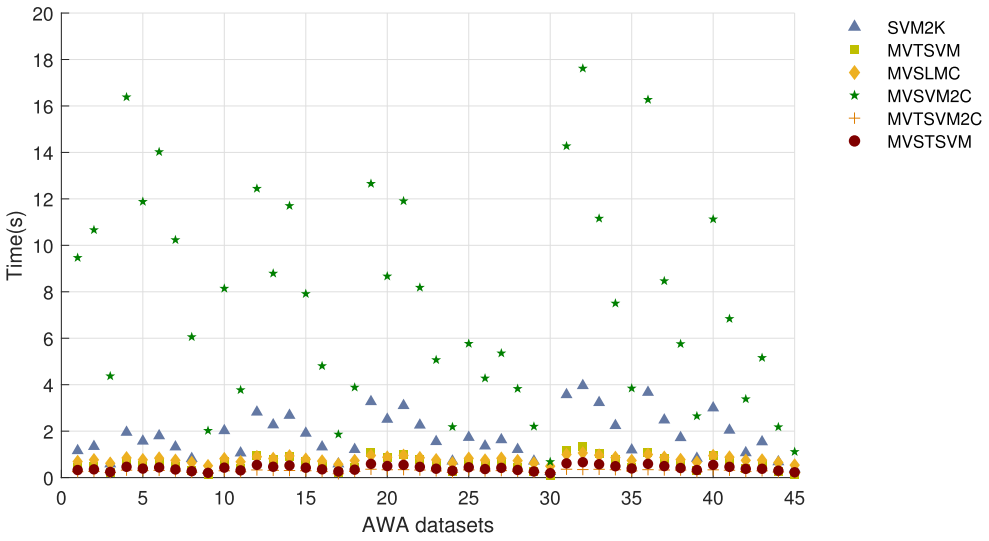


Fig. 5. The training time of all algorithms on AwA datasets.

Table 3
Training performance on two-view datasets.

Datasets	SVM2K		MvTSVM		MvSLMC		MvSVM2C		MvTSVM2C		MvSTSVM	
	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)
AWA_1	59.18 ± 4.65	1.16	66.60 ± 2.29	0.42	85.66 ± 2.08	0.69	78.24 ± 0.07	9.47	86.55 ± 1.78	0.26	**87.22 ± 2.18**	0.33
AWA_2	62.26 ± 2.37	1.34	64.83 ± 3.35	0.49	76.76 ± 2.26	0.77	69.52 ± 0.04	10.66	76.55 ± 3.33	0.27	**79.11 ± 1.71**	0.36
AWA_3	68.86 ± 5.42	0.59	56.95 ± 7.20	0.22	81.90 ± 6.33	0.62	60.37 ± 0.06	4.37	81.34 ± 6.79	0.22	**82.45 ± 3.99**	0.25
AWA_4	66.66 ± 0.85	1.95	66.68 ± 5.51	0.70	79.13 ± 2.93	0.85	77.27 ± 0.03	16.38	**81.19 ± 1.63**	0.32	80.44 ± 2.26	0.46
AWA_5	64.54 ± 2.77	1.57	60.66 ± 5.25	0.54	72.13 ± 1.92	0.77	69.48 ± 0.04	11.88	73.36 ± 1.88	0.30	**73.56 ± 1.93**	0.39
AWA_6	67.82 ± 3.41	1.80	63.59 ± 2.78	0.65	77.97 ± 1.43	0.84	76.04 ± 0.06	14.02	78.92 ± 3.78	0.31	**79.11 ± 3.60**	0.44
AWA_7	61.21 ± 3.56	1.32	60.78 ± 9.02	0.48	**73.50 ± 4.43**	0.74	69.84 ± 0.07	10.23	72.85 ± 2.96	0.28	73.48 ± 3.73	0.35
AWA_8	57.75 ± 3.62	0.82	55.49 ± 5.46	0.31	67.68 ± 1.80	0.65	62.37 ± 0.07	6.06	**68.21 ± 3.91**	0.23	66.18 ± 5.87	0.28
AWA_9	70.06 ± 6.34	0.31	66.08 ± 4.32	0.12	74.77 ± 6.41	0.51	68.23 ± 0.04	2.02	76.18 ± 2.57	0.20	**77.26 ± 2.37**	0.20
AWA_10	72.48 ± 3.73	2.02	66.79 ± 3.80	0.71	88.99 ± 2.25	0.81	82.75 ± 0.01	8.14	**89.17 ± 2.71**	0.29	88.07 ± 2.15	0.43
AWA_11	59.21 ± 1.70	1.06	69.47 ± 6.07	0.37	**93.48 ± 1.75**	0.68	82.74 ± 0.03	3.77	93.01 ± 1.15	0.25	93.01 ± 2.16	0.31
AWA_12	72.09 ± 9.16	2.82	75.37 ± 3.69	0.97	**93.97 ± 1.22**	0.90	88.58 ± 0.02	12.44	93.80 ± 1.46	0.33	93.48 ± 1.71	0.55
AWA_13	72.69 ± 4.65	2.27	67.74 ± 4.06	0.77	**88.48 ± 3.53**	0.82	83.51 ± 0.03	8.79	87.94 ± 4.03	0.31	88.30 ± 3.27	0.47
AWA_14	67.56 ± 3.57	2.68	64.22 ± 1.86	0.91	81.78 ± 3.27	0.93	78.76 ± 0.03	11.70	82.44 ± 2.56	0.30	**83.11 ± 2.24**	0.52
AWA_15	68.70 ± 4.11	1.91	68.15 ± 3.74	0.70	86.85 ± 2.65	0.79	84.44 ± 0.01	7.91	86.85 ± 4.36	0.30	**87.96 ± 2.36**	0.43
AWA_16	66.94 ± 6.09	1.32	63.53 ± 4.08	0.48	80.60 ± 1.71	0.71	81.44 ± 0.03	4.81	82.72 ± 2.81	0.26	**83.15 ± 3.53**	0.36
AWA_17	65.74 ± 7.44	0.59	76.49 ± 2.09	0.22	91.80 ± 4.02	0.57	81.03 ± 0.07	1.86	**92.08 ± 3.53**	0.17	90.10 ± 2.41	0.26
AWA_18	66.15 ± 4.84	1.21	64.59 ± 2.94	0.45	84.32 ± 5.00	0.73	81.87 ± 0.03	3.89	83.43 ± 4.54	0.27	**85.19 ± 4.41**	0.34
AWA_19	57.87 ± 4.25	3.27	70.60 ± 2.88	1.06	79.40 ± 1.67	0.95	71.38 ± 0.05	12.66	77.36 ± 2.46	0.36	**80.50 ± 2.95**	0.59
AWA_20	58.44 ± 4.26	2.51	64.40 ± 5.05	0.86	70.70 ± 3.78	0.88	58.78 ± 0.03	8.67	72.06 ± 1.66	0.34	**72.24 ± 2.66**	0.50
AWA_21	67.31 ± 2.12	3.10	68.27 ± 4.94	1.01	84.71 ± 4.65	0.95	71.83 ± 0.06	11.91	**86.16 ± 3.11**	0.34	84.70 ± 0.99	0.55
AWA_22	56.84 ± 4.95	2.26	64.82 ± 6.58	0.77	**71.92 ± 6.66**	0.85	54.35 ± 0.04	8.18	70.85 ± 6.24	0.33	71.75 ± 4.24	0.46
AWA_23	58.54 ± 2.15	1.55	61.40 ± 6.18	0.55	68.10 ± 3.27	0.77	60.57 ± 0.03	5.07	**69.71 ± 4.52**	0.29	68.30 ± 3.08	0.38
AWA_24	70.49 ± 13.85	0.71	75.80 ± 1.94	0.27	81.92 ± 1.09	0.63	77.13 ± 0.02	2.18	78.73 ± 2.01	0.22	**82.99 ± 2.79**	0.30
AWA_25	67.88 ± 0.53	1.73	72.31 ± 2.99	0.65	**82.50 ± 2.99**	0.83	79.42 ± 0.03	5.77	81.15 ± 4.79	0.33	81.73 ± 2.45	0.45
AWA_26	67.51 ± 3.81	1.36	64.97 ± 5.03	0.52	**79.83 ± 5.00**	0.75	71.96 ± 0.04	4.28	79.60 ± 4.34	0.28	79.19 ± 1.85	0.37
AWA_27	69.50 ± 2.75	1.63	68.91 ± 2.77	0.60	80.99 ± 5.35	0.83	61.39 ± 0.03	5.36	81.19 ± 3.13	0.30	**82.57 ± 4.78**	0.42
AWA_28	66.68 ± 4.80	1.21	63.53 ± 2.62	0.43	**79.87 ± 4.66**	0.73	73.84 ± 0.06	3.83	78.08 ± 3.41	0.27	**79.87 ± 2.24**	0.33
AWA_29	58.77 ± 3.46	0.71	62.22 ± 4.81	0.27	77.41 ± 3.84	0.62	52.92 ± 0.07	2.20	**78.46 ± 4.32**	0.23	77.67 ± 6.22	0.26
AWA_30	71.15 ± 7.81	0.26	66.54 ± 4.21	0.11	**78.85 ± 3.85**	0.49	72.31 ± 0.03	0.68	77.31 ± 6.58	0.21	78.46 ± 7.86	0.19
AWA_31	60.61 ± 1.58	3.57	67.94 ± 4.18	1.16	74.20 ± 1.98	0.99	65.34 ± 0.03	14.27	73.89 ± 1.47	0.36	**75.42 ± 2.98**	0.61
AWA_32	61.98 ± 5.22	3.96	73.59 ± 1.66	1.32	86.06 ± 3.00	1.06	70.97 ± 0.03	17.62	87.08 ± 3.79	0.35	**87.51 ± 2.17**	0.66
AWA_33	59.74 ± 1.82	3.23	66.09 ± 2.92	1.03	75.28 ± 2.82	0.97	60.54 ± 0.02	11.16	72.89 ± 3.40	0.36	**75.75 ± 2.81**	0.58
AWA_34	63.57 ± 1.16	2.24	68.21 ± 1.35	0.79	74.11 ± 2.09	0.86	67.68 ± 0.02	7.50	73.75 ± 1.49	0.34	**74.29 ± 2.71**	0.49
AWA_35	80.85 ± 3.12	1.19	78.38 ± 1.81	0.42	83.57 ± 3.82	0.74	80.86 ± 0.02	3.85	83.35 ± 3.88	0.29	**83.79 ± 1.99**	0.39
AWA_36	62.34 ± 4.89	3.67	68.44 ± 3.11	1.07	**78.59 ± 3.06**	0.99	71.72 ± 0.03	16.27	77.97 ± 3.38	0.34	77.97 ± 2.43	0.59
AWA_37	56.54 ± 4.22	2.48	60.13 ± 4.73	0.83	**62.37 ± 1.96**	0.87	57.74 ± 0.06	8.47	61.86 ± 3.75	0.33	61.69 ± 2.80	0.50
AWA_38	**60.86 ± 0.57**	1.72	54.60 ± 2.01	0.61	59.30 ± 0.49	0.78	60.08 ± 0.03	5.76	59.68 ± 4.10	0.32	56.37 ± 4.39	0.41
AWA_39	70.38 ± 3.86	0.83	68.61 ± 3.28	0.31	77.22 ± 1.27	0.66	76.71 ± 0.01	2.65	77.22 ± 2.00	0.25	**79.24 ± 1.44**	0.33
AWA_40	64.77 ± 4.33	3.00	64.28 ± 3.88	0.97	77.27 ± 2.60	0.95	70.29 ± 0.03	11.13	**79.70 ± 4.00**	0.33	**79.70 ± 2.68**	0.55
AWA_41	63.49 ± 1.99	2.04	66.97 ± 5.47	0.73	75.23 ± 3.61	0.88	70.28 ± 0.04	6.84	**75.41 ± 5.40**	0.32	74.50 ± 4.17	0.46
AWA_42	80.18 ± 3.51	1.07	78.32 ± 2.79	0.38	85.31 ± 4.18	0.74	79.46 ± 0.07	3.38	85.55 ± 6.46	0.26	**86.72 ± 2.38**	0.38
AWA_43	58.32 ± 1.64	1.54	59.35 ± 2.53	0.55	62.42 ± 2.04	0.75	58.74 ± 0.05	5.16	**64.29 ± 5.20**	0.29	61.80 ± 4.82	0.38
AWA_44	57.42 ± 4.07	0.68	70.07 ± 4.27	0.27	76.82 ± 1.43	0.65	75.47 ± 0.01	2.18	75.48 ± 1.90	0.24	**77.90 ± 2.74**	0.30
AWA_45	58.33 ± 5.40	0.38	61.00 ± 6.30	0.15	71.67 ± 2.64	0.54	70.00 ± 0.05	1.11	71.67 ± 2.64	0.21	**73.67 ± 3.21**	0.23
Harberman	73.53 ± 0.53	0.90	74.20 ± 4.93	0.35	73.53 ± 0.53	0.66	74.53 ± 0.01	0.75	**74.85 ± 2.28**	0.06	**74.85 ± 2.28**	0.16
Mushroom	59.85 ± 1.91	76.34	54.14 ± 0.07	22.76	58.76 ± 3.16	6.04	–	–	**60.81 ± 2.12**	0.89	60.17 ± 2.03	2.48
Avg.	64.97	3.32	66.30	1.07	78.03	0.89	69.98	7.33	78.14	0.30	**78.56**	0.45
Avg.Rank	5.23	4.94	5.17	2.96	2.28	3.89	4.38	5.98	2.17	1.09	**1.77**	2.15

• On Mushroom dataset, MvSVM2C is trained for more than 24 h, so the results of this item are not recorded in the table.

It needs to be clarified that 'Raw-MvSTSVM' means our model is conducted without safe screening process, while 'Screening ratio' refers to the percentage of screened Lagrange multipliers. The process of SSR-MvSTSVM is drawn in red line while Raw-MvSTSVM is in blue line. As shown, SSR-MvSTSVM consistently requires less training time than Raw-MvSTSVM because the safe screening process efficiently identifies and removes non-SVs from QPPs in advance. Meanwhile, as more multipliers are filtered out, the more running time is saved. That is to

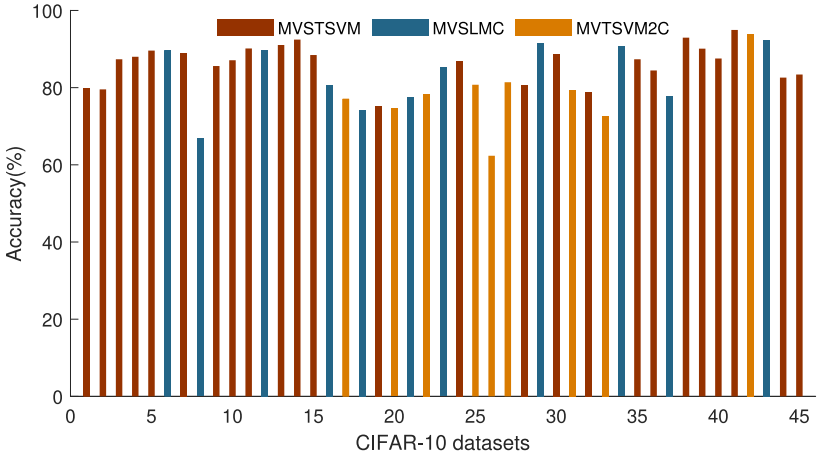


Fig. 6. Comparison of accuracy and time of multi-view algorithms on CIFAR-10.

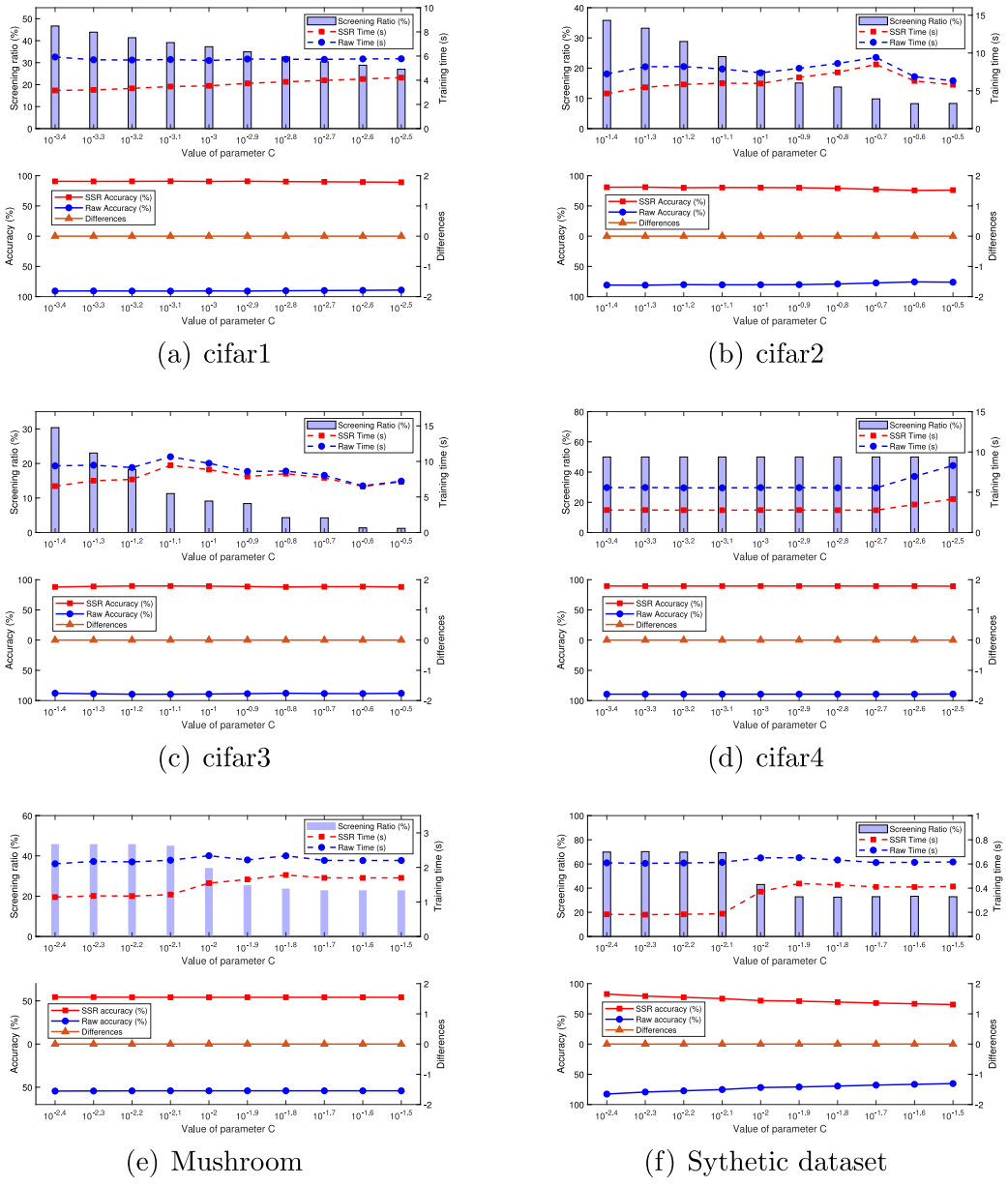


Fig. 7. The safe screening process on cifar1, cifar2, cifar3, cifar4, Mushroom and Sythetic dataset.

Table 4
AUC, F1-score and Precision on two-view datasets.

Datasets	SVM2K			MvTSVM			MvSLMC			MvSVM2C			MvTSVM2C			MvSTSVM		
	AUC	F1-score	Precision	AUC	F1-score	Precision	AUC	F1-score	Precision	AUC	F1-score	Precision	AUC	F1-score	Precision	AUC	F1-score	Precision
AWA_1	0.67	0.64	0.51	0.69	0.59	0.60	**0.93**	0.82	**0.88**	0.85	0.74	0.73	0.91	0.83	**0.88**	0.92	**0.84**	0.86
AWA_2	0.59	0.27	0.57	0.65	0.54	0.56	**0.82**	0.64	**0.84**	0.72	0.46	0.74	**0.82**	0.63	0.83	0.81	**0.70**	0.82
AWA_3	0.71	0.71	0.70	0.56	0.59	0.59	**0.88**	0.83	0.83	0.73	0.42	**0.88**	0.86	0.82	0.82	**0.88**	**0.84**	0.83
AWA_4	0.49	0.07	0.75	0.63	0.48	0.53	0.84	0.63	0.82	0.82	0.60	0.76	**0.87**	0.67	**0.85**	0.84	**0.68**	0.79
AWA_5	0.57	0.18	0.71	0.60	0.50	0.48	0.78	0.51	0.76	0.68	0.49	0.66	0.77	0.55	**0.77**	**0.79**	**0.59**	0.72
AWA_6	0.70	0.33	0.64	0.59	0.47	0.49	**0.86**	0.65	**0.75**	0.84	0.69	0.64	0.85	**0.70**	0.70	0.85	0.68	0.73
AWA_7	0.61	0.32	0.54	0.59	0.51	0.50	0.78	**0.62**	0.73	0.72	0.58	0.66	0.79	0.57	**0.78**	**0.80**	**0.62**	0.74
AWA_8	0.59	0.50	0.56	0.53	0.53	0.52	**0.72**	**0.67**	0.65	0.68	0.62	0.59	**0.72**	**0.67**	**0.66**	**0.72**	0.64	0.65
AWA_9	0.71	0.77	**0.79**	0.57	0.79	0.68	0.77	0.83	0.77	0.70	0.77	0.75	0.81	**0.84**	0.75	**0.82**	**0.84**	0.78
AWA_10	0.78	0.69	0.75	0.71	0.65	0.66	**0.97**	**0.88**	0.93	0.88	0.79	**0.96**	0.96	**0.88**	0.94	0.96	0.87	0.90
AWA_11	0.67	0.52	0.95	0.73	0.76	0.73	0.98	**0.95**	0.95	0.93	0.84	0.95	**0.98**	0.94	**0.96**	0.97	0.94	0.96
AWA_12	0.81	0.54	0.80	0.82	0.70	0.72	**0.98**	**0.93**	0.95	0.93	0.85	**0.97**	**0.98**	0.92	0.96	**0.98**	0.92	0.96
AWA_13	0.77	0.69	0.73	0.72	0.65	0.66	**0.94**	**0.87**	0.92	0.88	0.81	0.87	**0.94**	0.86	**0.94**	**0.94**	**0.87**	0.89
AWA_14	0.73	0.55	0.70	0.68	0.60	0.58	**0.91**	0.79	0.80	0.82	0.77	0.74	**0.91**	0.78	**0.86**	0.89	**0.80**	0.83
AWA_15	0.74	0.69	0.67	0.74	0.66	0.68	**0.95**	0.86	0.88	0.88	0.83	**0.93**	0.94	0.86	0.88	0.94	**0.87**	0.91
AWA_16	0.70	0.70	0.71	0.67	0.69	0.66	0.91	0.83	0.83	0.83	0.82	**0.91**	**0.92**	**0.85**	0.82	0.88	0.84	0.87
AWA_17	0.78	0.70	**0.95**	0.89	0.86	0.76	**0.96**	0.94	**0.95**	0.86	0.86	0.94	**0.96**	**0.95**	0.93	0.95	0.93	0.92
AWA_18	0.59	0.77	0.68	0.64	0.72	0.72	**0.93**	**0.89**	0.82	0.82	0.85	**0.86**	0.89	0.87	0.84	0.91	**0.89**	0.85
AWA_19	0.60	0.42	0.55	0.75	0.65	0.70	**0.88**	0.76	0.80	0.78	0.67	0.68	0.85	0.72	0.81	**0.88**	**0.77**	0.82
AWA_20	0.60	0.48	0.60	0.68	0.64	0.62	0.78	0.70	0.70	0.60	0.67	0.55	0.79	0.67	**0.79**	**0.80**	**0.71**	0.72
AWA_21	0.75	0.65	0.64	0.75	0.67	0.64	0.91	0.84	0.81	0.82	0.72	0.67	**0.93**	**0.85**	0.85	0.92	0.83	0.86
AWA_22	0.57	0.54	0.58	0.68	0.67	0.63	0.76	0.72	**0.72**	0.68	0.69	0.53	**0.77**	0.72	0.70	**0.77**	**0.73**	0.71
AWA_23	0.48	0.73	0.59	0.61	0.68	0.65	0.69	0.74	0.69	0.63	0.71	0.62	**0.73**	**0.77**	0.69	0.72	0.74	**0.70**
AWA_24	0.71	0.78	0.84	0.65	0.86	0.76	0.81	**0.89**	0.85	0.76	0.87	0.77	0.81	0.87	0.80	**0.83**	**0.89**	**0.86**
AWA_25	0.55	0.01	0.20	0.70	0.51	0.59	**0.88**	0.67	**0.85**	0.84	0.64	0.74	0.84	0.66	0.78	0.85	**0.70**	0.74
AWA_26	0.67	0.34	0.59	0.64	0.51	0.51	0.86	**0.68**	0.78	0.81	0.64	0.59	0.86	0.64	**0.85**	**0.88**	0.66	0.79
AWA_27	0.69	0.22	0.71	0.70	0.54	0.53	**0.89**	0.71	0.71	0.76	0.60	0.46	0.87	0.71	0.72	0.88	**0.72**	**0.77**
AWA_28	0.68	0.46	0.64	0.65	0.54	0.51	0.85	**0.70**	0.79	0.78	0.65	0.66	0.84	0.68	0.75	**0.86**	**0.70**	**0.81**
AWA_29	0.59	0.59	0.53	0.65	0.57	0.58	0.84	**0.75**	0.74	0.79	0.66	0.49	**0.85**	**0.75**	**0.78**	0.84	0.73	**0.78**
AWA_30	0.77	0.76	**0.82**	0.66	0.77	0.69	0.83	**0.84**	0.81	0.80	0.81	0.72	0.82	0.83	0.80	**0.85**	**0.84**	**0.82**
AWA_31	0.63	0.64	0.63	0.72	0.72	0.68	0.81	0.77	**0.74**	0.69	0.73	0.63	0.82	**0.79**	0.70	**0.83**	0.78	**0.74**
AWA_32	0.64	0.57	0.67	0.79	0.76	0.71	0.93	0.87	0.83	0.84	0.77	0.65	0.93	**0.88**	0.85	**0.94**	**0.88**	**0.87**
AWA_33	0.60	0.65	0.63	0.72	0.72	0.67	**0.83**	0.79	**0.76**	0.61	0.72	0.60	**0.83**	**0.80**	0.69	0.82	0.79	**0.76**
AWA_34	0.42	0.77	0.64	0.65	0.77	0.71	**0.82**	**0.82**	0.72	0.75	0.79	0.67	0.77	0.81	**0.74**	0.79	**0.82**	**0.74**
AWA_35	0.64	0.89	0.82	0.57	0.88	0.80	0.83	**0.90**	**0.86**	0.80	0.89	0.81	**0.85**	**0.90**	0.85	**0.85**	**0.90**	0.85
AWA_36	0.64	0.58	0.60	0.71	0.67	0.67	**0.85**	**0.77**	0.78	0.78	0.73	0.66	**0.85**	0.76	0.78	**0.85**	0.75	**0.80**
AWA_37	0.56	0.63	0.56	0.61	0.62	0.62	**0.66**	0.65	0.63	0.60	0.61	0.58	**0.66**	**0.68**	0.60	**0.66**	0.63	**0.64**
AWA_38	0.57	0.75	0.60	0.52	0.64	**0.61**	0.55	**0.74**	0.59	0.55	**0.74**	0.60	**0.58**	0.73	**0.61**	0.57	0.66	**0.61**
AWA_39	0.53	0.81	0.79	0.59	0.80	0.79	0.70	0.87	0.77	0.63	0.87	0.77	0.68	0.87	0.78	**0.71**	**0.88**	**0.80**
AWA_40	0.70	0.70	0.66	0.67	0.68	0.67	**0.87**	0.79	**0.81**	0.77	0.70	0.79	**0.87**	**0.82**	0.80	0.86	**0.82**	**0.81**
AWA_41	0.61	0.76	0.64	0.67	0.74	0.73	**0.82**	**0.82**	0.75	0.73	0.75	0.78	0.81	0.81	0.77	0.80	0.79	**0.79**
AWA_42	0.78	0.87	0.88	0.73	0.88	0.80	**0.89**	0.91	0.90	0.71	0.86	**0.92**	0.87	0.91	0.89	**0.89**	**0.92**	0.87
AWA_43	0.53	**0.73**	0.58	0.58	0.65	0.64	0.62	0.69	**0.66**	0.59	0.72	0.59	**0.65**	**0.73**	0.65	0.61	0.70	0.64
AWA_44	0.60	0.68	0.77	0.61	0.80	**0.79**	0.73	0.86	0.77	0.58	0.86	0.75	0.68	0.86	0.78	**0.76**	**0.87**	0.78
AWA_45	0.56	0.66	0.75	0.58	0.72	0.72	0.72	**0.83**	0.72	0.60	0.79	**0.77**	0.70	0.82	0.73	**0.73**	**0.83**	0.74
Harberman	0.67	**0.85**	0.74	0.60	0.84	**0.78**	0.65	**0.85**	0.74	0.58	**0.85**	0.74	**0.69**	**0.85**	0.75	0.66	**0.85**	0.75
mushroom	**0.64**	**0.59**	0.56	**0.64**	0.50	0.46	0.63	0.47	0.54	–	–	–	0.63	0.49	0.61	**0.64**	0.45	**0.61**
Avg.	0.64	0.59	0.67	0.66	0.67	0.65	**0.83**	**0.78**	0.78	0.75	0.73	0.72	0.82	**0.78**	**0.79**	**0.83**	**0.78**	**0.79**

say, the percentage of screened Lagrange multipliers roughly reflects the percentage improvement in the training speed of SSR-MvSTSVM.

In the second subplot, we focus on the optimal classification accuracy of both SSR-MvSTSVM and Raw-MvSTSVM. The left y-axis represents the optimal accuracies achieved by both methods, while the right y-axis illustrates the difference in their accuracies. We can find that the classification accuracies of SSR-MvSTSVM and Raw-MvSTSVM are identical across all datasets, which confirms the safety of our screening method. This consistency in accuracy demonstrates that SSR-MvSTSVM successfully reduces computational burden without compromising the performance of the model.

5.5. Parameter analysis

5.5.1. Parameter analysis on model accuracy

During the numerical experiment, the optimal accuracy is often affected by parameter settings. To study the influence of $C = C_+ = C_-, D = D_+ = D_-, \lambda = \eta$ in MvSTSVM, we conduct experiments on Mushroom dataset.

We first fix λ and vary C, D from 10^{-3} to 10^3 with exponential term step of 1 and work out the highest accuracy on Mushroom dataset. The details are visualized in Fig. 8. The grid is linked to different parameter groups. The color of each scatter point indicates the corresponding highest accuracy. We can see that when λ, D are fixed, the optimal accuracy is affected by the penalty parameter C. While the value of C is smaller, the color of the point is lighter, indicating that the accuracy is higher. Otherwise, the result is the opposite.

Next, we fix C and vary D, λ the same way as above in Fig. 9. Our MvSTSVM achieves the principle of complementarity by introducing the view weight controlled by parameter λ. When D is fixed and C is not too large, we can conclude that the overall performance of MvSTSVM is getting better with λ changing from 10^{-3} to 10^3. This is because when the value of λ is relatively small, our model is similar to a single-view model. The model contains less perspective information, so the training result is poor. However, when C is set at 10 or 10^2, the accuracy goes down or stabilizes instead with the increasing λ. In this case, for a larger λ, the view weights are close to equally divided, therefore the model accuracy tends to decrease or remain constant.

Moreover, Figs. 8 and 9 both depict the variation of D. For fixed C and λ, the corresponding accuracy changes very little when D varies.

5.5.2. Parameter analysis on safe screening

To solve the dual formulation more efficiently, we derive SSR to accelerate the training session. In this process, C and D are two variables that may affect the effectiveness of screening. To learn how the screening process is influenced by these two variables, we first fix λ, D and vary C from 10^{-3} to 10^3 with exponential term step of 1 and record the screening rate on synthetic dataset. The "screening rate" here is equal to the ratio of the screened sample numbers to the total number of training samples. It is worth noting that SSR does not perform in the model with parameter $C = 10^{-3}$ because the first MvSTSVM should be solved without SSR before parameter screening.

The variation of the screening rate is depicted in Fig. 10. The graph shows that the screening rate has something to do with C and D. When D, λ is fixed, our method obtains a high screening rate when $C = 10^{-1}$

Table 5
Experiment results on multi-view datasets.

Datasets	MvSLMC		MvTSVM2C		MvSTSVM	
	Accuracy (%)	CPU Time (s)	Accuracy (%)	CPU Time (s)	Accuracy (%)	CPU Time (s)
cifar1	79.55 ± 0.93	5.13	78.60 ± 1.71	3.15	**79.90 ± 1.10**	1.43
cifar2	78.95 ± 1.27	4.99	77.80 ± 1.15	3.07	**79.55 ± 1.34**	1.34
cifar3	87.00 ± 1.70	4.91	85.80 ± 1.84	3.28	**87.35 ± 2.01**	1.53
cifar4	87.45 ± 0.99	4.90	87.10 ± 1.70	3.25	**88.00 ± 1.80**	1.49
cifar5	89.05 ± 1.66	4.96	88.90 ± 1.81	3.21	**89.60 ± 1.65**	1.49
cifar6	**89.70 ± 2.13**	4.97	89.15 ± 2.14	3.24	89.50 ± 1.68	1.51
cifar7	88.85 ± 1.86	4.99	88.15 ± 1.41	3.30	**89.00 ± 1.69**	1.48
cifar8	**66.85 ± 1.29**	4.98	65.50 ± 2.26	3.16	66.05 ± 1.32	1.37
cifar9	85.05 ± 2.28	5.04	85.35 ± 1.27	3.33	**85.60 ± 1.39**	1.49
cifar10	86.00 ± 2.16	5.01	85.55 ± 2.71	3.30	**87.10 ± 1.43**	1.49
cifar11	89.60 ± 1.97	5.13	89.50 ± 1.45	3.31	**90.15 ± 1.78**	1.47
cifar12	**89.75 ± 1.37**	5.04	89.70 ± 1.62	3.18	89.55 ± 1.19	1.47
cifar13	90.65 ± 1.76	5.07	90.80 ± 1.32	3.20	**91.05 ± 1.51**	1.45
cifar14	91.75 ± 1.21	4.99	92.05 ± 1.22	3.11	**92.45 ± 1.43**	1.45
cifar15	87.35 ± 2.28	5.01	87.55 ± 1.57	3.21	**88.45 ± 1.76**	1.47
cifar16	**80.75 ± 1.57**	5.18	79.80 ± 2.01	3.05	80.65 ± 1.72	1.32
cifar17	77.05 ± 1.79	5.12	**77.10 ± 1.26**	3.27	76.95 ± 1.64	1.49
cifar18	**74.25 ± 1.16**	5.02	72.45 ± 3.25	3.16	73.90 ± 1.70	1.35
cifar19	74.95 ± 1.48	5.14	74.65 ± 2.44	3.31	**75.30 ± 2.84**	1.54
cifar20	73.80 ± 1.24	5.04	**74.60 ± 1.66**	3.17	74.55 ± 1.15	1.49
cifar21	**77.65 ± 1.34**	5.07	77.60 ± 1.70	3.30	77.55 ± 1.52	1.54
cifar22	78.10 ± 1.21	5.08	**78.30 ± 0.74**	3.34	78.15 ± 1.40	1.51
cifar23	**85.20 ± 3.02**	4.90	83.75 ± 1.76	3.16	85.10 ± 2.54	1.46
cifar24	85.85 ± 2.61	4.91	86.45 ± 1.48	3.32	**86.85 ± 2.40**	1.45
cifar25	80.35 ± 1.83	5.04	**80.75 ± 0.79**	3.17	80.55 ± 1.05	1.51
cifar26	62.30 ± 2.94	5.02	**62.35 ± 2.59**	3.07	61.30 ± 2.48	1.34
cifar27	81.15 ± 1.55	5.06	**81.40 ± 1.65**	3.24	81.25 ± 1.22	1.55
cifar28	80.55 ± 1.87	5.10	80.30 ± 1.45	3.29	**80.60 ± 1.88**	1.53
cifar29	**91.55 ± 0.94**	4.91	89.80 ± 1.04	3.19	91.35 ± 1.01	1.46
cifar30	88.65 ± 1.02	5.12	88.40 ± 1.04	3.27	**88.80 ± 0.89**	1.45
cifar31	78.95 ± 1.81	5.05	**79.40 ± 2.12**	3.18	79.20 ± 1.79	1.55
cifar32	78.55 ± 2.50	5.10	78.30 ± 3.93	3.34	**78.80 ± 2.81**	1.50
cifar33	72.60 ± 2.00	5.10	**72.65 ± 2.52**	3.19	72.25 ± 3.62	1.44
cifar34	**90.65 ± 1.61**	4.98	89.75 ± 2.18	3.17	90.25 ± 1.90	1.44
cifar35	86.85 ± 1.35	4.96	86.75 ± 2.28	3.14	**87.35 ± 2.04**	1.45
cifar36	84.00 ± 1.29	5.02	83.95 ± 1.44	3.30	**84.45 ± 2.62**	1.49
cifar37	**77.75 ± 1.25**	7.99	77.70 ± 1.41	3.23	**77.75 ± 1.51**	1.50
cifar38	92.75 ± 0.79	8.02	92.15 ± 1.39	3.14	**92.95 ± 1.08**	1.44
cifar39	89.55 ± 0.89	8.16	89.60 ± 1.18	3.14	**90.10 ± 1.10**	1.45
cifar40	87.15 ± 0.84	5.04	87.15 ± 1.10	3.29	**87.55 ± 1.27**	1.44
cifar41	94.90 ± 1.85	4.96	94.45 ± 1.60	3.19	**94.95 ± 1.78**	1.47
cifar42	93.30 ± 1.52	4.89	**93.90 ± 1.56**	3.11	93.75 ± 0.83	1.40
cifar43	**92.35 ± 1.26**	5.00	91.70 ± 1.37	3.17	**92.35 ± 0.96**	1.47
cifar44	82.35 ± 1.26	4.88	82.25 ± 1.45	3.12	**82.60 ± 2.30**	1.46
cifar45	83.25 ± 1.79	5.03	83.05 ± 1.93	3.18	**83.40 ± 1.93**	1.48
sythetic	99.25 ± 1.47	3.11	98.47 ± 0.97	0.65	**100.00 ± 0.76**	0.82
Avg.	84.00	5.18	83.72	3.16	**84.21**	1.45
Avg.Rank	2.02	3	2.43	2	1.54	**1**

or $C = 10^{-2}$. The highest screening rate is nearly 80%. In addition, for the same parameter C, the screening rate varies with different D. In a word, the performance of MvSTSVM is closely associated with the parameter settings. Consequently, the values of them must be carefully chosen to achieve optimal results.

5.6. Experiments on parameter intervals

In the search sequence $\{10^i | i = -3 : d : 3\}$ of parameter C, the parameter interval d has something to do with the highest accuracy we obtained. In the above experiments, we have acquired the optimal accuracy when d is equal to 1. To further improve the classification performance of MvSTSVM, we narrow the search range and reduce the parameter interval to $d = 0.02$ for training. For instance, if $C = 10^i$ corresponds to the best performance in the previous experiment, the current search sequence of C will be set in $\{10^j \mid j = (i-1) : 0.02 : (i+1)\}$ to obtain higher accuracy. In this part, we take the first six dataset from CIFAR-10 as examples and adopt the safe screening rules to accelerate the solution. The experiment result is listed in Table 7. "RawAcc" and "RawTime" refer to the best accuracy and average time respectively without the process of SSR. "Ratio" is defined as the percentage of screened Lagrange multipliers, which reflects percentage increase in training time.

From the results, we can see that the optimal accuracy increases with refining parameter C. Moreover, the training process spends less time in selecting one parameter when "$d = 0.02$" than that in "$d = 1$".

In Table 7, it is obvious that "RawAcc" and "SSRAcc" are the same, which can prove that our proposed SSR-MvSTSVM is secure. Meanwhile, due to the acceleration of SSR-MvSTSVM, the average time during the training process decreases. In addition, the efficiency of SSR in speeding up the training process is linked to the size of the parameter interval. When d is equal to 0.02, the screening ratio is higher than training with $d = 1$.

To further explore the relationship between "d" and "Ratio", we conduct an experiment on the same dataset when d is set to $\{0.02, 0.04, 0.06, 0.08, 0.1\}$. The results are drawn in Fig. 11, which demonstrates the screening rate on different parameter intervals. On all

Table 6
AUC,F1-score and Precision on multi-view datasets.

Datasets	MvSLMC			MvTSVM2C			MvSTSVM		
	AUC	Precision	F1-score	AUC	Precision	F1-score	AUC	Precision	F1-score
cifar1	0.87	**0.83**	0.78	0.86	0.81	0.78	**0.88**	0.79	**0.80**
cifar2	0.85	0.78	0.79	0.83	0.77	0.78	**0.86**	**0.79**	**0.80**
cifar3	0.93	**0.89**	0.87	0.92	0.86	0.86	**0.94**	0.88	**0.87**
cifar4	**0.95**	**0.88**	0.87	0.94	0.87	0.87	0.94	0.87	**0.88**
cifar5	**0.95**	**0.93**	0.89	**0.95**	0.91	0.89	**0.95**	0.89	**0.90**
cifar6	**0.96**	**0.91**	**0.90**	0.95	0.91	0.89	**0.96**	0.91	0.89
cifar7	**0.95**	**0.91**	0.89	**0.95**	0.90	0.88	**0.95**	0.89	**0.89**
cifar8	0.71	**0.75**	0.62	0.70	0.66	0.65	**0.72**	0.67	**0.65**
cifar9	**0.93**	**0.86**	0.85	0.92	0.86	0.85	**0.93**	0.86	**0.86**
cifar10	0.92	0.82	0.87	0.92	0.84	0.86	**0.94**	**0.89**	**0.87**
cifar11	0.95	0.89	0.90	0.95	0.89	0.90	**0.96**	**0.89**	**0.90**
cifar12	**0.96**	0.89	**0.90**	0.95	0.89	0.90	**0.96**	**0.89**	0.90
cifar13	0.96	**0.91**	0.91	0.96	0.90	0.91	**0.97**	0.91	**0.91**
cifar14	**0.97**	0.91	0.92	**0.97**	**0.92**	0.92	**0.97**	0.92	**0.93**
cifar15	0.93	0.87	0.87	0.93	0.88	0.87	**0.95**	**0.89**	**0.88**
cifar16	**0.88**	0.79	**0.81**	0.87	**0.81**	0.79	**0.88**	0.80	0.81
cifar17	**0.85**	**0.77**	**0.77**	**0.85**	0.76	0.77	**0.85**	0.77	0.77
cifar18	0.81	**0.78**	0.72	0.80	0.72	0.73	**0.82**	0.75	**0.73**
cifar19	**0.83**	0.76	0.74	**0.83**	0.76	0.74	**0.83**	**0.76**	**0.75**
cifar20	0.80	**0.77**	0.72	0.81	0.75	**0.75**	**0.82**	0.76	0.74
cifar21	**0.85**	**0.80**	0.77	**0.85**	0.77	**0.78**	**0.85**	0.77	0.78
cifar22	**0.86**	0.78	0.78	**0.86**	0.78	**0.79**	**0.86**	**0.78**	0.78
cifar23	**0.93**	**0.89**	0.84	0.91	0.85	0.83	**0.93**	0.87	**0.85**
cifar24	0.93	0.89	0.85	0.93	0.84	**0.87**	**0.94**	**0.89**	0.87
cifar25	**0.88**	0.79	0.81	**0.88**	**0.80**	**0.81**	**0.88**	0.80	0.81
cifar26	**0.67**	0.61	**0.64**	**0.67**	**0.63**	0.62	0.65	0.62	0.61
cifar27	**0.88**	0.82	0.81	**0.88**	**0.82**	0.81	**0.88**	0.81	**0.81**
cifar28	**0.88**	0.80	**0.81**	**0.88**	0.80	0.80	**0.88**	**0.81**	0.81
cifar29	**0.97**	**0.92**	**0.92**	**0.97**	0.90	0.90	**0.97**	0.91	0.91
cifar30	**0.95**	0.90	0.88	**0.95**	**0.91**	0.89	**0.95**	0.88	**0.89**
cifar31	0.87	0.78	0.79	**0.88**	0.79	**0.80**	0.87	**0.79**	0.79
cifar32	**0.87**	**0.80**	0.78	0.85	0.79	0.78	**0.87**	0.78	**0.79**
cifar33	**0.79**	0.73	**0.72**	**0.79**	**0.73**	0.72	0.78	0.72	0.72
cifar34	0.96	**0.92**	**0.90**	0.96	0.91	0.90	**0.97**	0.92	0.90
cifar35	**0.94**	**0.89**	0.87	**0.94**	0.86	0.87	**0.94**	0.89	**0.87**
cifar36	**0.91**	**0.85**	0.84	**0.91**	0.84	0.84	**0.91**	0.84	**0.85**
cifar37	**0.86**	0.78	**0.78**	0.85	0.78	0.78	**0.86**	**0.78**	0.78
cifar38	**0.98**	**0.93**	0.93	**0.98**	0.92	0.92	**0.98**	0.93	**0.93**
cifar39	**0.96**	0.90	0.90	**0.96**	0.88	0.90	**0.96**	**0.90**	**0.90**
cifar40	**0.94**	0.87	0.87	**0.94**	0.87	0.87	**0.94**	**0.88**	**0.88**
cifar41	**0.99**	**0.95**	0.95	0.98	0.95	0.94	**0.99**	0.95	**0.95**
cifar42	**0.98**	0.94	0.93	**0.98**	0.94	**0.94**	**0.98**	**0.95**	0.94
cifar43	**0.97**	**0.93**	0.92	**0.97**	0.92	0.92	**0.97**	0.92	**0.92**
cifar44	**0.91**	**0.83**	0.82	0.90	0.82	0.82	**0.91**	0.81	**0.83**
cifar45	0.90	0.84	0.83	0.90	**0.85**	0.83	**0.91**	0.84	**0.83**
sythetic	0.99	0.98	0.99	0.99	0.98	0.98	**1.00**	**1.00**	**1.00**
Avg.	0.904	**0.849**	0.837	0.901	0.839	0.836	**0.906**	0.844	**0.842**

Table 7
Experiment results on parameter intervals learning.

Datasets	Intervals	RawAcc (%)	RawTime (s)	SSRAcc (%)	SSRTime (s)	Ratio
cifar1	$d = 1$	79.90 ± 1.10	1.43	79.90 ± 1.10	1.31	0.09
cifar2		79.55 ± 1.34	1.34	79.55 ± 1.34	1.19	0.11
cifar3		87.35 ± 2.01	1.53	87.35 ± 2.01	1.38	0.09
cifar4		88.00 ± 1.80	1.49	88.00 ± 1.80	1.36	0.09
cifar5		89.60 ± 1.65	1.49	89.60 ± 1.65	1.37	0.08
cifar6		89.50 ± 1.68	1.51	89.50 ± 1.68	1.40	0.07
cifar1	$d = 0.02$	79.90 ± 0.95	1.07	79.90 ± 0.95	0.58	0.46
cifar2		79.55 ± 1.29	1.00	79.55 ± 1.29	0.67	0.33
cifar3		87.50 ± 2.22	1.15	87.50 ± 2.22	0.77	0.33
cifar4		87.80 ± 1.57	1.12	87.80 ± 1.57	0.58	0.49
cifar5		89.60 ± 1.68	1.12	89.60 ± 1.68	0.61	0.45
cifar6		89.75 ± 1.84	1.13	89.75 ± 1.84	0.70	0.38

six datasets, we can conclude consistently that the screening efficiency of SSR-MvSTSVM decreases gradually when d increases.

Interval d has an impact that cannot be ignored on our experiment results. But something should be noted that the total time of the parameter selection will increase if d is smaller. Therefore, an appropriate parameter interval is necessary in our experiments.

6. Conclusion

This article introduces a new algorithm named MvSTSVM, which satisfies consensus and complementarity principles simultaneously. MvSTSVM integrates structural information from inter-class and intra-class data by employing a clustering method to maintain consistency

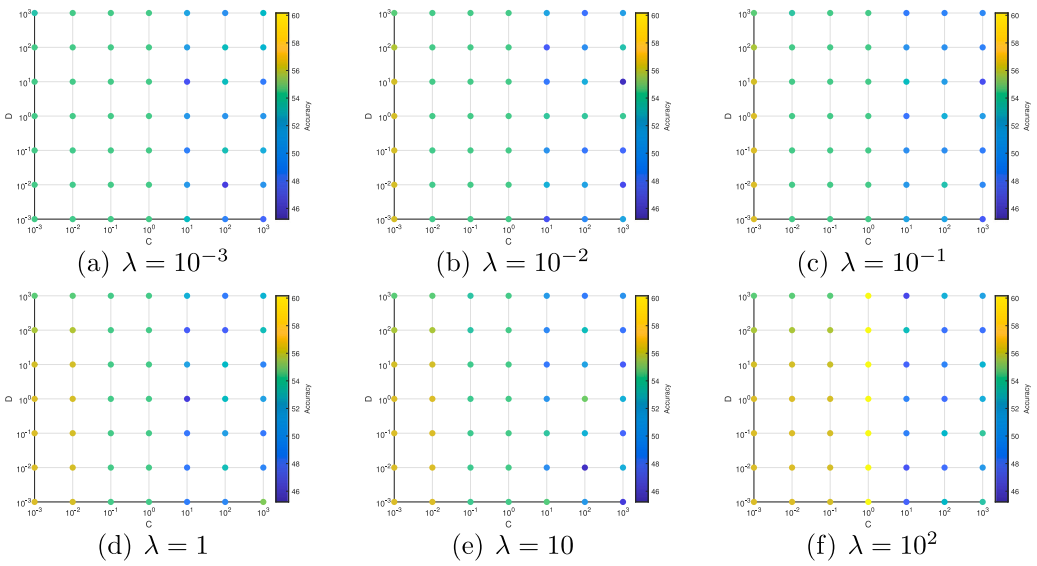


Fig. 8. Accuracies under different C, D on dataset Mushroom.

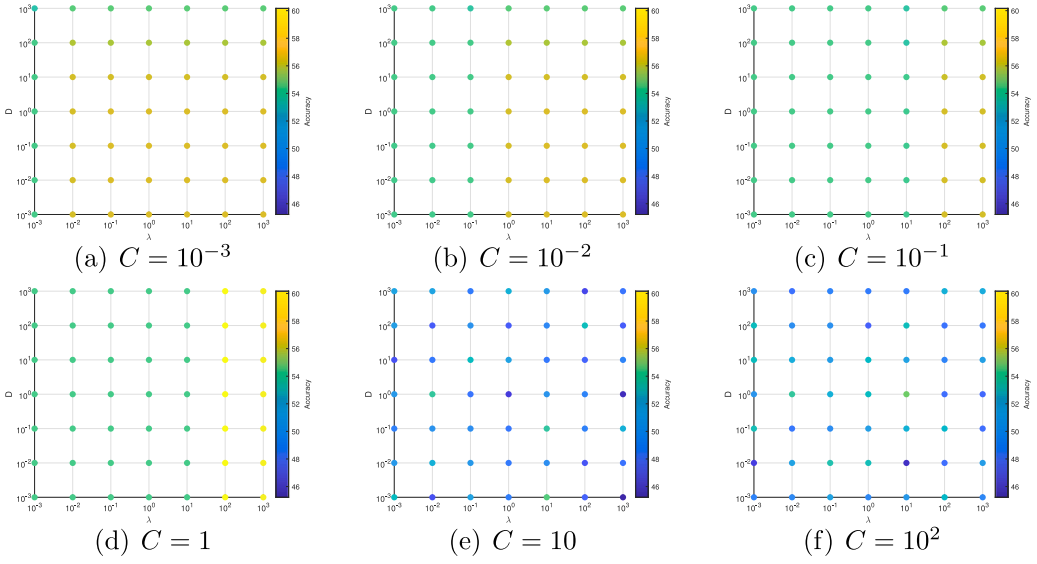


Fig. 9. Accuracies under different D, λ on dataset Mushroom.

information. Additionally, it implements complementarity by introducing perspective weights into multiple views. To improve the training efficiency, an iterative algorithm is utilized for the solution of optimization problems. Furthermore, SSR for MvSTSVM is established to accelerate the tuning process without changing the solution of the original problem. A series of statistics experiments are conducted to demonstrate the efficiency of MvSTSVM and the safety of the screening rules. In the future, we will work on more accelerating algorithms to improve computational efficiency.

CRediT authorship contribution statement

Qianfei Liu: Methodology, Software, Validation, Formal Analysis, Investigation, Data curation, Writing – original draft, Visualization. **Chen Chen:** Software, Validation, Writing – review & editing. **Ting Huang:** Formal analysis, Writing – review & editing. **Yan Meng:** Writing – review & editing. **Huiru Wang:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

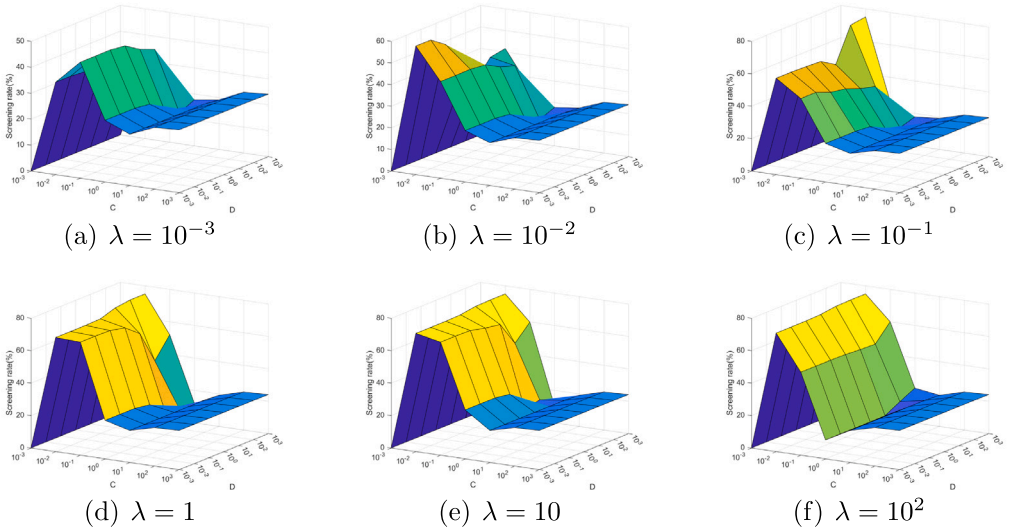


Fig. 10. Screening rate under different parameter settings on synthetic dataset.

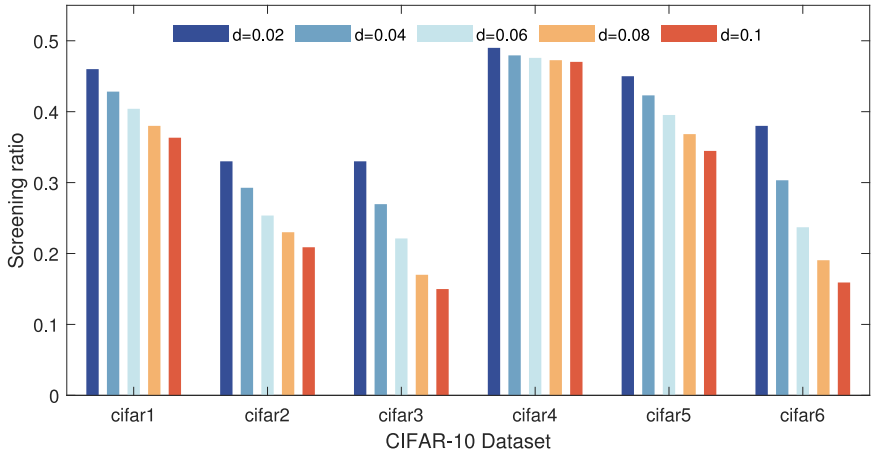


Fig. 11. Safe screening ratio for different parameter intervals from cifar1 to cifar6.

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